Chapter 64

Exploring the Fusion Interface

This chapter provides an orientation on the Fusion user interface, providing a quick tour of what tools are available, where to find things, and how the different panels fit together to help you build and refine compositions in this powerful node-based environment.

Contents

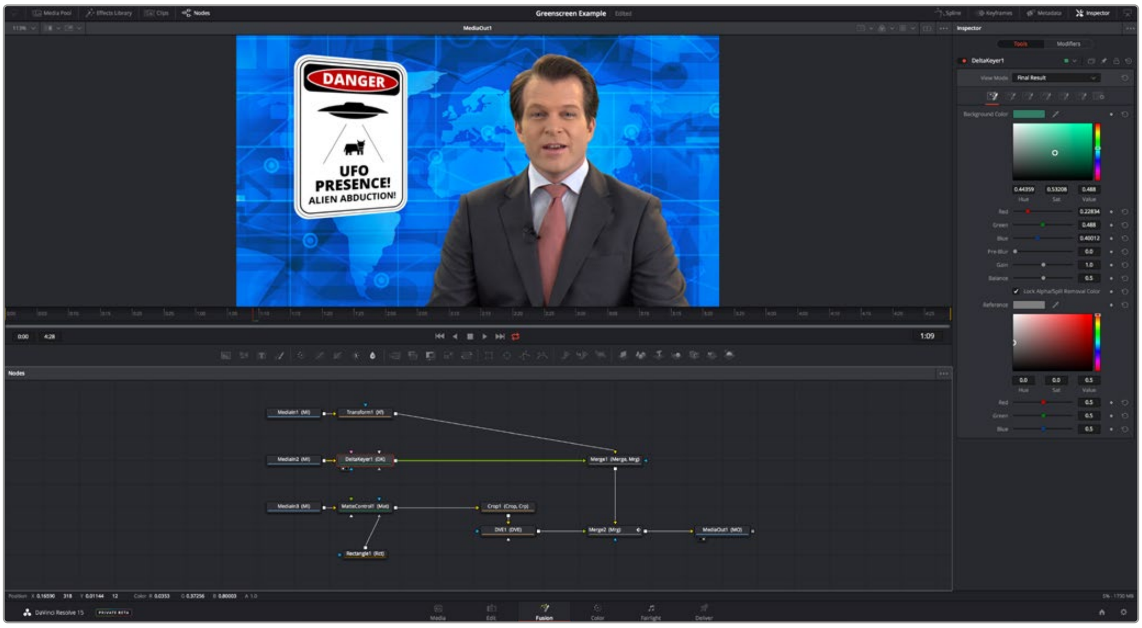
FUSION

The Fusion User Interface

If you open up everything at once, Fusion is divided into four principal regions designed to help you make fast work of node-based compositing. The viewer(s) are at the top, the work area is at the bottom, the Inspector is at the right, and the Effects Library is the area found at the left. In DaVinci Resolve's Fusion page, the Effects Library shares space with the Media Pool. All these panels work together to let you add effects, paint to correct issues, create motion graphics or title sequences, or build sophisticated 3D and multi-layered composites.

The Fusion user interface shown completely

However, Fusion doesn't have to be that complicated, and in truth, you can work very nicely with only the viewer, Node Editor, and Inspector open for a simplified experience.



A simplified set of Fusion controls for everyday working

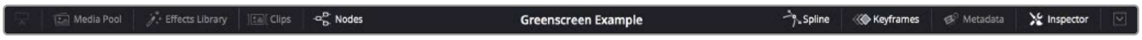
The Work Area

You probably won't see the term "the work area" used much, in favor of the specific panels within the work area that you'll be using. Still, the area referred to as the work area is the region at the bottom half of the Fusion user interface, within which you can expose the three main panels used to construct compositions and edit animations in Fusion. These are the Node Editor, the Spline Editor, and the Keyframes Editor. By default, the Node Editor is the first thing you'll see, and the main area you'll be working within, but it can sit side-by-side with the Spline Editor and Keyframes Editor as necessary. You can make more horizontal room on your display for these three panels by putting the Effects Library and Inspector into a half-height mode, if necessary.

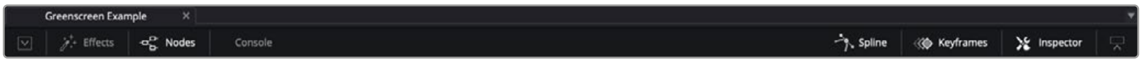
The work area showing the Node Editor, the Spline Editor, and the Keyframes Editor

Interface Toolbar

At the very top of Fusion is a toolbar with buttons that let you show and hide different parts of the user interface (UI). Buttons with labels identify which parts of the UI can be shown or hidden. In DaVinci Resolve's Fusion page, if you right-click anywhere within this toolbar, you have the option of displaying this bar with or without text labels.



The UI toolbar of the Fusion page



The UI toolbar of Fusion Studio

These buttons are as follows, from left to right:

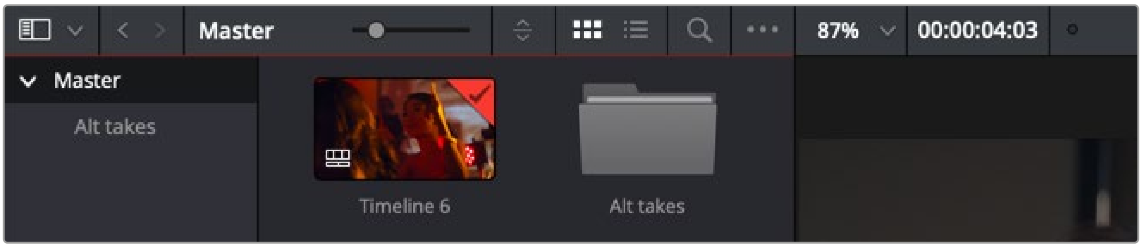
- **Media Pool/Effects Library Full Height:** Lets you set the area used by the Media Pool (DaVinci Resolve only) and/or Effects Library to take up the full height of your display, giving you more area for browsing at the expense of a narrower Node Editor and viewer area. At half-height, the Media Pool/Templates/Effects Library are restricted to the top half of the UI along with the viewers (you can only show one at a time), and the Node Editor takes up the full width of your display.
- **Media Pool:** (DaVinci Resolve only): Shows and hides the Media Pool, from which you can drag additional clips into the Node Editor to use them in your Fusion page composition.
- **Effects Library:** Opens or hides the repository of all node tools available to use in Fusion. From here, you can click nodes to add them after the currently selected node in the Node Editor, or you can drag and drop nodes to any part of the node tree you like.
- **Clips:** (DaVinci Resolve only): Opens and closes the Thumbnail timeline, which lets you navigate your program, create and manage multiple versions of compositions, and reset the current composition.
- **Nodes:** Opens and closes the Node Editor, where you build and edit your compositions.

- **Console (Fusion Studio only):** The Console is a window in which you can see the error, log, script, and input messages that may explain something Fusion is trying to do in greater detail. The Console is also where you can read FusionScript outputs, or input FusionScripts directly.
- **Spline:** Opens and closes the Spline Editor, where you can edit the curves that interpolate keyframe animations to customize and perfect their timing. Each keyframed parameter appears hierarchically within the effect in which it appears in a list to the left.
- **Keyframes:** Opens and closes the Keyframes Editor, which shows each clip and effects node in your Fusion composition as a layer. You can use the Keyframes Editor to edit and adjust the timing of keyframes that have been added to various effects in your composition. You can also use the Keyframes Editor to slide the relative timing of clips that have been added to Fusion, as well as to trim their In and Out points. A spreadsheet can be shown and hidden within which you can numerically edit keyframe values for selected effects.
- **Metadata (DaVinci Resolve only):** Hides or shows the Metadata Editor, which lets you read and edit the available clip and project metadata associated with any piece of media within a composite.
- **Inspector:** Shows or hides the Inspector, which shows you all the editable parameters and controls that correspond to selected nodes in the Node Editor. You can show the parameters for multiple nodes at once, and even pin the parameters of nodes you need to continue editing so that they're displayed even if those nodes aren't selected.
- **Inspector Height:** Lets you open the Inspector to be half height (the height of the viewer area) or full height (the height of your entire display). Half height allows more room for the Node Editor, Spline Editor, and/or Keyframes Editor, but full height lets you simultaneously edit more node parameters or have enough room to display the parameters of multiple nodes at once.

Choosing Which Panel Has Focus

Whenever you click somewhere on the Fusion interface using the pointer or using a keyboard shortcut to "select" a particular panel, you give that panel of the user interface "focus." A panel with focus captures specific keyboard shortcuts to do something within that panel, as opposed to doing something elsewhere in the interface.

To make it easier to keep track of which panel has focus, a highlight appears at the top edge of whichever panel has focus. In DaVinci Resolve, you must turn on "Show focus indicators in the User Interface" in the UI Settings panel of the User Preferences to see the highlight.



The focus indicator shown at the top edge of the Media Pool, shown next to a viewer that doesn't have focus

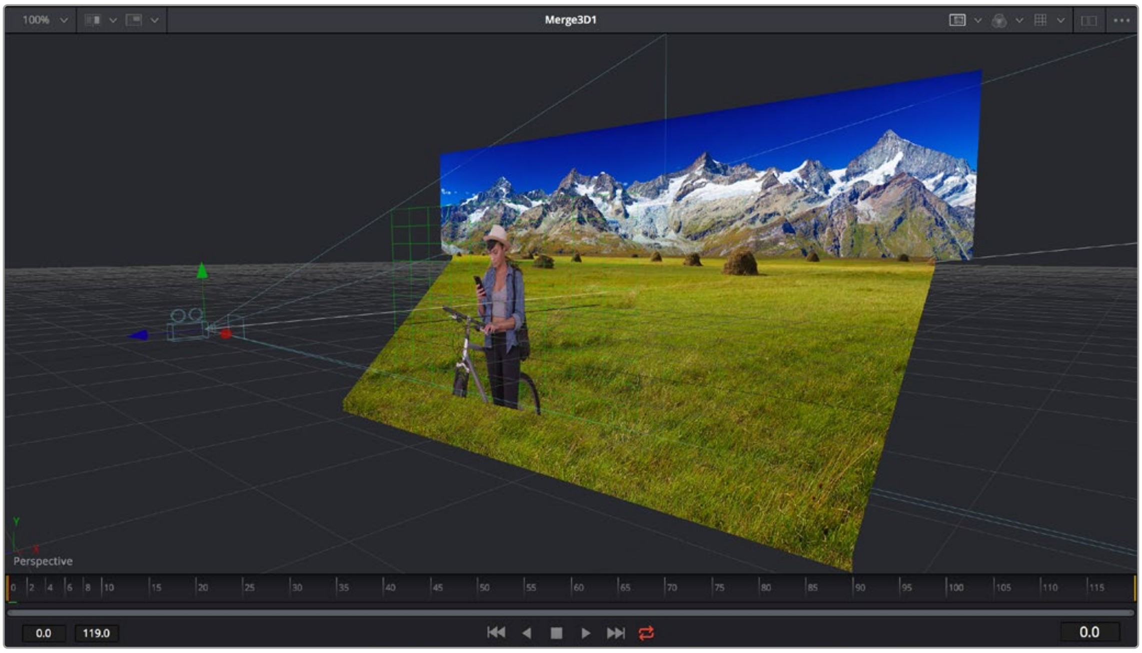
FUSION

Viewers

The viewer area displays either one or two viewers at the top of the Fusion page, and this is determined via the Viewer button at the far right of the Viewer title bar. Each viewer can show a single node's output from anywhere in the node tree. You assign which node is displayed in which viewer. This makes it easy to load separate nodes into each viewer for comparison. For example, you can load a Keyer node into the left viewer and the final composite into the right viewer, so you can see the image you're adjusting and the final result at the same time.

Dual viewers let you edit an upstream node in one while seeing its effect on the overall composition in the other.

Ordinarily, each viewer shows 2D nodes from your composition as a single image. However, when you're viewing a 3D node, you have the option to set that viewer to one of several 3D views. A perspective view gives you a repositionable stage on which to arrange the elements of the world you're creating. Alternatively, a quad view lets you see your composition from four angles, making it easier to arrange and edit objects and layers within the XYZ axes of the 3D space in which you're working.



Loading a 3D node into a viewer switches on a Perspective view

FUSION

TIP: In Perspective view, you can hold down the middle and right mouse buttons, then drag in the viewer to pivot the view around the center of the world. All other methods of navigating viewers work the same.

The viewers have a variety of capabilities you can use to compare and evaluate images. This section provides a short overview of viewer capabilities to get you started.

Zooming and Panning into Viewers

There are standardized methods of zooming into and panning around viewers when you need a closer look at the situation. These methods also work with the Node Editor, Spline Editor, and Keyframes Editor.

Methods of panning viewers:

- Middle-click and drag to pan around the viewer.
- Hold down Shift and Command and drag the viewer to pan.
- Drag with two fingers on a track pad to pan.

Methods of scaling viewers:

- Click a viewer, and press the equals key (=) to zoom in, and the minus key (-) to zoom out.
- Press the middle and left mouse buttons simultaneously and drag left or right to resize the viewer.
- Hold down the Command key and use your pointer's scroll control to zoom in and out of the viewer.
- Hold down the middle mouse button, and then click the left mouse button to zoom in, or click the right button to zoom out. The scaling uses a fixed amount, centered on the position of the cursor.
- Click a viewer and press Command-1 to resize the image in the viewer to 100 percent.
- Click a viewer and press Command-2 to resize the image in the viewer to 200 percent.
- Click a viewer and press F to reset the image in the viewer to fit the viewer.
- Click the Scale viewer menu and choose Fit or a percentage.
- Right-click on a viewer and choose an option from the Scale submenu of the contextual menu. This includes a Custom Scale command that lets you type your own scale percentage.
- Hold down the Command key and drag with two fingers on a track pad to zoom in and out of the viewer.

Methods of spinning 3D viewers:

- In 3D Perspective view, hold down the middle and right mouse button, then drag to spin the stage around.
- In 3D Perspective view, hold down the Shift key and drag with two fingers on a track pad to spin the stage around.

Loading Nodes Into Viewers

When you first open the Fusion page in DaVinci Resolve, the output of the current empty composition (the MediaOut1 node) is usually showing in viewer 2. If you're in Dual-viewer mode, viewer 1 remains empty until you assign a node to one of them.

When using Fusion Studio, nothing is loaded into either of the viewers until you assign a node to one of them.

To load specific nodes into specific viewers:

- Hover the pointer over a node, and click one of two buttons that appear at the bottom left of the node.
- Click once to select a node, and press 1 (for the left viewer) or 2 (for the right viewer).
- Right-click a node and choose View On > None/Left View/Right View in the contextual menu.
- Right-click the control header of a node in the Inspector, and choose View On > None/Left View/Right View from the contextual menu.
- Drag a node and drop it over the viewer you'd like to load it into (this is great for tablet users).

When a node is being viewed, a View Indicator button appears at the bottom left. This is the same control that appears when you hover the pointer over a node. Not only does this control let you know which nodes are loaded into which viewer, but they also expose little round buttons for switching between viewers.

Viewer assignment buttons at the bottom of nodes indicate when they're being viewed.

Clearing Viewers

To clear an image from a viewer, click in the viewer to make it active; a thin red highlight is displayed at the top of the active viewer. With the viewer active, press the ` (accent) key. This key is usually found to the left of the 1 key on U.S. keyboards. The fastest way to remove all the images from all the viewers is to make sure none of the viewers is the active panel, and then press the Tilde key.

You can also select the node that is currently showing in the viewer, and press the viewer number again (1 or 2 respectively) to clear the viewer.

Viewer Controls

A series of buttons and pop-up menus in the viewer's title bar provides several quick ways of customizing the viewer display.

Controls in the viewer title bar

- **Zoom menu:** Lets you zoom in on the image in the viewer to get a closer look, or zoom out to get more room around the edges of the frame for rotoscoping or positioning different layers. Choose Fit to automatically fit the overall image to the available dimensions of the viewer.
- **Split Wipe button and A/B Buffer menu:** You can actually load two nodes into a single viewer using that viewer's A/B buffers by choosing a buffer from the menu and loading a node into the viewer. Turning on the Split Wipe button (press Forward Slash) shows a split wipe between the two buffers, which can be dragged left or right via the handle of the onscreen control, or rotated by dragging anywhere on the dividing line on the onscreen control. Alternatively, you can switch between each full-screen buffer to compare them (or to dismiss a split-screen) by pressing Comma (A buffer) and Period (B buffer).

- **SubView type:** (These aren't available in 3D viewers.) Clicking the icon itself enables or disables the current "SubView" option you've selected, while using the menu lets you choose which SubView is enabled. This menu serves one of two purposes. When displaying ordinary 2D nodes, it lets you open up SubViews, which are viewer "accessories" within a little pane that can be used to evaluate images in different ways. These include an Image Navigator (for navigating when zoomed far into an image), Magnifier, 2D viewer (a mini-view of the image), 3D Histogram scope, Color Inspector, Histogram scope, Image Info tooltip, Metadata tooltip, Vectorscope, or Waveform scope. The Swap option (Shift-V) lets you switch what's displayed in the viewer with what's being displayed in the Accessory pane. When displaying 3D nodes, this button lets you have access to an additional mini 3D viewer.
- **Node name:** The name of the currently viewed node is displayed at the center of the viewer's title bar.
- **ROI controls:** Clicking the icon itself enables or disables RoI (Region of Interest) limiting in the viewer, while using the menu lets you choose the region of the RoI. RoI lets you define the region of the viewer in which pixels actually need to be updated. When a node renders, it intersects the current RoI with the current Domain of Definition (DoD) to determine what pixels should be affected. When enabled, you can position a rectangle to restrict rendering to a small region of the image, which can significantly speed up performance when you're working on very high resolution or complex compositions. Auto (the default) sets the region to whatever is visible at the current zoom/pan level in the viewer. Choosing Set lets you draw a custom region within the frame by dragging a rectangle that defaults to the size of the viewer, which is resizable by dragging the corners or sides of the onscreen control. Choosing Lock prevents changes from being made to the current RoI. Choosing Reset resets the RoI to the whole viewer.
- **Color controls:** Lets you choose which color and/or image channels to display in the viewer. Clicking the icon itself toggles between Color (RGB) and Alpha, the two most common things you want to see (pressing C or A also toggles between Color and Alpha). Opening the menu displays every possible channel that can be displayed for the currently viewed node, commonly including RGB, Red, Green, Blue, and Alpha (available from the keyboard by pressing R, G, B, or A). For certain media and nodes, additional auxiliary channels are available to be viewed, including Z-depth, Object ID, Material ID, XYZ Normals, and so on.
- **Viewer LUT:** Clicking the icon itself toggles LUT (LookUp Table) display on or off, while the menu lets you choose which of the many available color space conversions to apply to the viewer. The top options let you choose Fusion controls that can be customized via the Edit item at the top of this menu. The rest of this menu shows all LUTs installed in the LUT directory to use for viewing.

  By default, when using DaVinci Resolve, the viewers in the Fusion page show you the image prior to any grading done in the Color page, since the Fusion page comes before the Color page in the DaVinci Resolve image processing pipeline. When you're working on clips that have been converted to linear color space for compositing, it is desirable to composite and make adjustments to the image relative to a normalized version of the image that appears close to what the final will be. Enabling the LUT display lets you do this as a preview, without permanently applying color adjustments to the image.
- **Option menu:** This menu contains various settings that pertain to the viewers in Fusion.

  Snap to Pixel: When drawing or adjusting a polyline mask or spline, the control points will snap to pixel locations.

  Show Controls: Toggles whatever onscreen controls are visible for the currently selected node.

  Region: Provides all the settings for the Region of Interest in the viewer.

  Smooth Resize: This option uses a smoother bilinear interpolated resizing method when zooming into an image in the viewer; otherwise, scaling uses the nearest neighbor method and shows noticeable aliasing artifacts. However, this is more useful when you zoom in at a pixel level since there is no interpolation.

FUSION

Show Square Pixels: Overrides the auto aspect correction when using formats with non-square pixels.

Checker Underlay: Toggles a checkerboard underlay that makes it easy to see areas of transparency.

Normalized Color Range: Allows for the visualization of brightness values outside of the normal viewing range, particularly when working with floating-point images or auxiliary channels.

Gain/Gamma: Exposes a simple pair of Gain and Gamma sliders that let you adjust the viewer's brightness.

360 View: Used to properly display spherical imagery in a variety of formats, selectable from this submenu.

Stereo: Used to properly display stereoscopic imagery in a variety of formats, selectable from this submenu.

Time Ruler and Transport Controls

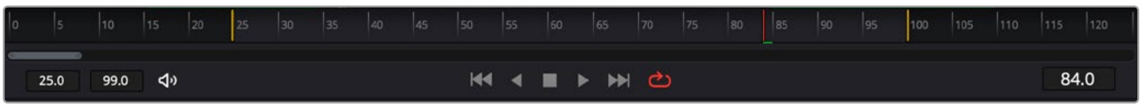
The Time Ruler, located beneath the viewer area, is based on the total duration of the composition. What it represents depends on which version of Fusion you're using:

- For DaVinci Resolve users, the duration displayed in the Time Ruler range depends on what's currently selected in the Edit or Cut page timeline.
- In Fusion Studio, the Time Ruler depends on the Global Start and End values set in the Fusion Studio Preferences > Defaults.

The transport controls under the Time Ruler include playback controls, audio monitoring, as well as number fields for the composition duration and playback range. Additional controls enable motion blur and proxy settings.

Time Ruler Controls in the Fusion Page

If you've selected a single clip in the Edit or Cut page Timeline, then the global range shown in the Time Ruler is based on the total source duration for that clip. You cannot move the playhead outside the global range. The yellow lines, called the render range, identify the current In and Out points for the clip and are the only frames visible in the Fusion page. All frames outside of this range constitute the unused head and tail handles of that source clip.



The Time Ruler displaying ranges for a clip in the Timeline via yellow marks (the playhead is red)

If you've created a Fusion clip or a compound clip, then the "working range" reflects the entire duration of that clip.

The Time Ruler displaying ranges for a Fusion clip in the Timeline

Render Range

The render range determines the range of frames that are visible in the Fusion page and that are used for interactive playback, disk caches, and previews. Frames outside the default render range are not visible in the Fusion page and are not rendered or played.

You can modify the duration of the render range for preview and playback only. Making the range shorter or longer does not trim the clip in the Edit or Cut page Timelines.

You can change the render range in the Time Ruler by doing one of the following:

- Hold down the Command key and drag a new range within the Time Ruler.
- Drag either the start or end yellow line to modify the start or end of the range.
- Right-click within the Time Ruler and choose Set Render Range from the contextual menu.
- Enter new ranges in the Range In and Out fields to the left of the transport controls.
- Drag a node from the Node Editor to the Time Ruler to set the range to the duration of that node.

You can return the render range to the In and Out points of the timeline clip by doing one of the following.

- Right-click within the Time Ruler and choose Auto Render Range.
- Click back in the Edit or Cut page, and then return to the Fusion page.

Time Ruler Controls in Fusion Studio

The Time Ruler, located beneath the viewer area, displays two different frame ranges: one for the entire composition, called the global range, and the other called the render range, which determines what to render and what to cache in memory for previews. The global start and end range takes up the entire Time Ruler and sets the total duration of a composition. You cannot move the playhead outside the global range.

The Time Ruler displaying ranges for a clip in the Timeline via yellow marks (the playhead is red)

Global Start and End Range

The global start and end range is simply the total duration of the current composition.

You can change the global range by doing one of the following:

- To change the global range for all new compositions, choose Fusion Studio > Preferences on macOS or File > Preferences on Windows or Linux. In the Global and Default Settings panel, enter a new range in the Global range fields.
- To change the Global range for the current composition, enter a new range in the Global Start and End fields to the left of the transport controls.
- Dragging a node from the Node Editor to the Time Ruler automatically sets the Global and Render Range to the extent of the node.

Render Range

The render range determines the range of frames used for interactive playback, disk caches, and previews. Frames outside the render range are not rendered or played, although you can still drag the playhead to these frames to see the unused frames.

To preview or render a specific range of a composition, you can modify the render range in a variety of ways.

You can set the render range in the Time Ruler by doing one of the following:

- Hold down the Command key and drag a new range within the Time Ruler.
- Right-click within the Time Ruler and choose Set Render Range from the contextual menu to set the Render Range based on the selected Node's duration.
- Enter new ranges in the Range In and Out fields to the left of the transport controls.
- Drag a node from the Node Editor to the Time Ruler to set the range to the duration of that node.

The Playhead

A red playhead within the Time Ruler indicates the currently viewed frame. Clicking anywhere within the Time Ruler jumps the playhead to that frame, and dragging within the Time Ruler drags the playhead within the available duration of that clip or composition.

Zoom and Scroll Bar

A two-handled gray scroll bar lets you zoom into the range shown by the Time Ruler, which is useful if you're using a very large Global range such that the Render range is a tiny sliver in the Time Ruler. Dragging the left or right handles of this bar zooms relative to the opposite handle, enlarging the width of each displayed frame. Once you've zoomed in, you can drag the scroll bar left or right to scroll through the composition.

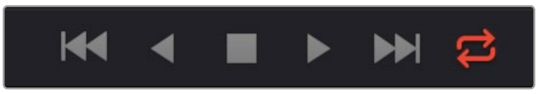
TIP: Holding the middle mouse button and dragging in the Time Ruler lets you scroll the visible range.

Transport Controls in the Fusion Page

The Transport Controls in DaVinci Resolve's Fusion page include buttons that control playback as well as time fields on the left side for setting the render range and the current time on the right side. Additional controls are available in the right-click menu.

Controlling Playback

There are six transport controls underneath the Time Ruler in the Fusion page. These buttons include Composition First Frame, Play Reverse, Stop, Play Forward, Composition Last Frame, and Loop.



The Fusion page controls for playback

Navigation Shortcuts

Many standard transport control keyboard shortcuts you may be familiar with work in Fusion, but some are specific to Fusion's particular needs.

To move the playhead in the Time Ruler using the keyboard, do one of the following:

Spacebar: Toggles forward playback on and off.

JKL: Basic JKL playback is supported, including J to play backward, K to stop, and L to play forward.

Back Arrow: Moves 1 frame backward.

Forward Arrow: Moves 1 frame forward.

Shift-Back Arrow: Moves to the clip's Global Start frame.

Shift-Forward Arrow: Moves to the clip's Global End frame.

Command-Back Arrow: Jumps to the Render Range In point.

Command-Forward Arrow: Jumps to the Render Range Out point.

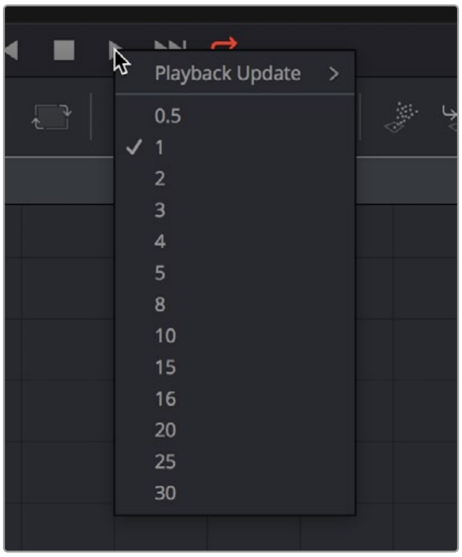
Real-Time Playback Not Guaranteed

Because many of the effects you can create in the Fusion page are processor-intensive, there is no guarantee of real-time playback at your project's full frame rate unless you've cached your composition first (discussed later).

Frame Increment Options

Right-clicking either the Play Reverse or Play Forward buttons opens a contextual menu. This menu contains options to set a frame increment value, which lets you use a keyboard shortcut to move the playhead in sub-frame or multi-frame increments.

Moving the playhead in multi-frame increments can be useful when rotoscoping. Moving the playhead in sub-frame increments can be useful when rotoscoping or inspecting interlaced frames one field at a time (0.5 of a frame).



Right-click the Play Forward or Play Backward buttons to choose a frame increment in which to move the playhead.

FUSION

Looping Options

The Loop button can be toggled to enable or disable looping during playback. You can right-click this button to choose the looping method that's used:

Playback Loop: The playhead plays to the end of the Time Ruler and starts from the beginning again.

Ping-pong Loop: When the playhead reaches the end of the Time Ruler, playback reverses until the playhead reaches the beginning of the Time Ruler, and then continues to ping-pong back and forth.

Render Range Fields

The two time fields on the left side of the transport controls are used to modify the Render Range. You can enter time values in frames, or click and drag inside the fields to modify the In and Out of the render range for previews and caching.

The Render Start and Render End time fields

Audio Monitoring

Playing a composition in DaVinci Resolve's Fusion page will play the audio from the Edit or Cut page Timeline. You can choose to hear the audio or mute it using the Audio toolbar button to the left of the transport controls. The audio waveforms are displayed in the Keyframes Editor to assist in the timing of your animations.

TIP: If the Mute button is enabled on any Timeline tracks, audio from those tracks will not be heard in Fusion.

For Fusion Studio, audio can be loaded using the Loader node's Audio tab. The audio functionality is included in Fusion Studio for scratch track (aligning effects to audio and clip timing) purposes. Final renders should almost always be performed without audio. Audio can be heard if it is brought in through a Loader node.

To hear the audio from a specific Loader node:

Right-click over the Speaker icon and choose the file name that contains the audio you want to hear.

Audio Toolbar Button

The Audio button in the toolbar is a toggle that can be used to enable or mute audio playback associated with the clip. Additionally, right-clicking this button displays a contextual menu that can be used to select a MediaIn node in the Fusion page or an external WAV file in Fusion Studio.

The Current Time Field

The Current Time field at the right of the transport controls displays the frame number for the playhead position, which corresponds to the frame seen in the viewer. Clicking and dragging in this field scrubs the playhead position back and forth. However, you can also enter time values into this field to move the playhead by specific amounts.

When setting ranges and entering frame numbers to move to a specific frame, numbers can be entered in sub-frame increments. You can set a range to be –145.6 to 451.75 or set the playhead to 115.22. This can be very helpful when animating parameters because you can set keyframes where they actually need to occur, rather than on a frame boundary, so you get more natural animation. Having sub-frame time lets you use time remapping nodes or just scale keyframes in the Spline view and maintain precision.

The Fusion Page Viewer Quality and Proxy Options

Right-clicking anywhere in the transport control area other than over the Play Forward/Play Reverse buttons lets you turn on and off Fusion quality controls. You can either enable high-quality playback at the expense of more significant processing times or enter various proxy modes that temporarily lower the display quality of your composition to speed processing as you work.

Rendering for final output is always done at the highest quality, regardless of these settings.

High Quality

As you build a composition, often the quality of the displayed image is less important than the speed at which you can work. The High Quality setting gives you the option to either display images with faster interactivity or at final render quality. When you turn off High Quality, complex and time-consuming operations such as area sampling, anti-aliasing, and interpolation are skipped to render the image to the viewer more quickly. Enabling High Quality forces a full-quality render to the viewer that's identical to what is output during final delivery.

Motion Blur

The Motion Blur button is a global setting. Turning off Motion Blur temporarily disables motion blur throughout the composition, regardless of any individual nodes for which it's enabled. This can significantly speed up renders to the viewer. Individual nodes must first have motion blur enabled before this button has any effect.

Proxy

The Proxy setting is a draft mode used to speed processing while you're building your composite. Turning on Proxy reduces the resolution of the images that are rendered to the viewer, speeding render times by causing only one out of every x pixels to be processed, rather than processing every pixel. The value of x is decided by adjusting a slider in the Proxy section in the Fusion > Fusion Settings > General panel.

Auto Proxy

The Auto Proxy setting is a draft mode used to speed processing while you're building your composite. Turning on Auto Proxy reduces the resolution of the image while you click and drag to adjust a parameter. Once you release that control, the image snaps back to its original resolution. This lets you adjust processor-intensive operations more smoothly, without the wait for every frame to render at full quality causing jerkiness. You can set the auto proxy ratio by adjusting a slider in the Proxy section of the Fusion > Fusion Settings > General panel.

Selective Updates

When working in Fusion, only the tools needed to display the images in the viewer are updated. The Selective Update options select the mode used during previews and final renders.

The options are available in the Proxy section of the Fusion > Fusion Settings > General panel.

The three options are:

- **Update All (All):** Forces all the nodes in the current node tree to render. This is primarily used when you want to update all the thumbnails displayed in the Node Editor.
- **Selective (Some):** Causes only nodes that directly contribute to the current image to be rendered. So named because only selective nodes are rendered. This is the default setting.
- **No Update (None):** Prevents rendering altogether, which can be handy for making many changes to a slow-to-render composition.

Transport Controls in Fusion Studio

The transport controls in Fusion Studio include buttons to control playback, time fields on the left side for setting the global range and render range, and a Render button to initiate rendering of the composite. There are also controls on the right side for proxy and motion blur. The time field on the far right is used for the current time.

Fusion Studio transport controls

Controlling Playback

There are eight transport controls underneath the Time Ruler in Fusion Studio. These buttons include Composition First Frame, Step Backward, Play Reverse, Stop, Play Forward, Step Forward, Composition Last Frame, and Loop.

Fusion Studio transport controls

Navigation Shortcuts

Many standard transport control keyboard shortcuts you may be familiar with work in Fusion, but there are some keyboard shortcuts specific to Fusion's particular needs.

To move the playhead in the Time Ruler using the keyboard, do one of the following:

- **Spacebar:** Toggles forward playback on and off.
- **JKL:** Basic JKL playback is supported, including J to play backward, K to stop, and L to play forward.
- **Back Arrow:** Moves 1 frame backward.
- **Forward Arrow:** Moves 1 frame forward.

- **Shift-Back Arrow:** Moves to the clip's Global End frame.
- **Shift-Forward Arrow:** Moves to the clip's Global Start frame.
- **Command-Back Arrow:** Jumps to the Render Range In point.
- **Command-Forward Arrow:** Jumps to the Render Range Out point.

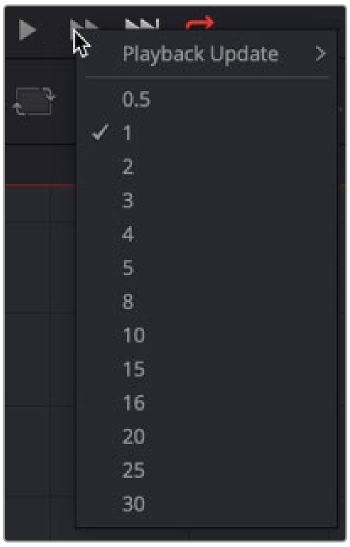
Real-Time Playback Not Guaranteed

Because many of the effects you can create in the Fusion page are processor-intensive, there is no guarantee of real-time playback at your project's full frame rate, unless you've cached your composition first. For more information, see the "Fusion RAM Cache for Playback" section later in this chapter.

Frame Increment Options

Right-clicking the Step Backward, Play Reverse, Play Forward, or Step Forward buttons opens a drop-down menu with options to set a frame increment value. Selecting a frame number from the menu lets you move the playhead in sub-frame or multi-frame increments whenever you use a keyboard shortcut or press the Step Forward/Backward buttons.

Moving the playhead in multi-frame increments can be useful when rotoscoping. Moving the playhead in sub-frame increments can be useful when rotoscoping or inspecting interlaced frames one field at a time (0.5 of a frame).



Right-click the Step Forward or Step Backward buttons to choose a frame increment in which to move the playhead.

Looping Options

The Loop button can be toggled to enable or disable looping during playback. You can right-click this button to choose the looping method that's used:

- **Playback Loop:** The playhead plays to the end of the Time Ruler and starts from the beginning again.
- **Ping-pong Loop:** When the playhead reaches the end of the Time Ruler, playback reverses until the playhead reaches the beginning of the Time Ruler, and then continues to ping-pong back and forth.

Range Fields

The four time fields on the left side of the transport controls are used to quickly modify the global range and render range in Fusion Studio.

Global Time Fields to the left of the transport controls

Audio

The Audio button is a toggle that mutes or enables any audio associated with the clip. Additionally, right-clicking on this button displays a drop-down menu that can be used to select a WAV file, which can be played along with the composition, and to assign an offset to the audio playback.

Render

Clicking the Render button in the transport controls displays the composition's Render Settings dialog. This dialog is used to configure the render options and initiate rendering of any Saver nodes in the composition. Shift-clicking on the button skips the dialog, using default render values (full resolution, high quality, motion blur enabled).

The Current Time

The Current Time field at the right of the transport controls shows the frame at the position of the playhead, which corresponds to the frame seen in the viewer. However, you can also enter time values into this field to move the playhead by specific amounts.

When setting ranges and entering frame numbers to move to a specific frame, numbers can be entered in sub-frame increments. You can set a range to be –145.6 to 451.75 or set the playhead to 115.22. This can be very helpful when animating parameters because you can set keyframes where they actually need to occur, rather than on a frame boundary, so you get more natural animation. Having sub-frame time lets you use time remapping nodes or just scale keyframes in the Spline view and maintain precision.

> **NOTE:** Many fields in Fusion can evaluate mathematical expressions that you type into them. For example, typing 2 + 4 into most fields results in the value 6.0 being entered. Because Feet + Frames uses the + symbol as a separator symbol rather than a mathematical symbol, the Current Time field will not correctly evaluate mathematical expressions that use the + symbol, even when the display format is set to Frames mode.

Fusion Studio Viewer Quality and Proxy Options

Five buttons along the right side of the transport controls let you either enable high-quality playback at the expense of greater processing times, or enter various proxy modes that temporarily lower the display quality in order to speed processing as you work.

Rendering for final output is always done at the highest quality, regardless of these settings.

Five buttons control the viewer quality, motion blur, proxy options, and image-processing update settings.

HiQ

As you build a composition, often the quality of the displayed image is less important than the speed at which you can work. The High Quality setting gives you the option to either display images with faster interactivity or at final render quality. When you turn off High Quality, complex and time-consuming operations such as area sampling, anti-aliasing, and interpolation are skipped to render the image to the viewer more quickly. Enabling High Quality forces a full-quality render to the viewer that's identical to what will be output during final delivery.

MB

The Motion Blur button is a global setting. Turning off Motion Blur temporarily disables motion blur throughout the composition, regardless of any individual nodes for which it's enabled. This can significantly speed up renders to the viewer. Individual nodes must first have motion blur enabled before this button has any effect.

Prx

A draft mode to speed processing while you're building your composite. Turning on Proxy reduces the resolution of the images that are rendered to the viewer, speeding render times by causing only one out of every x pixels to be processed, rather than processing every pixel. The value of x is decided by adjusting a slider in the General panel of the Fusion Preferences, found under the Fusion menu on macOS or the File menu on Windows and Linux.

Aprx

A draft mode to speed processing while you're building your composite. Turning on Auto Proxy reduces the resolution of the image while you click and drag to adjust a parameter. Once you release that control, the image snaps back to its original resolution. This lets you adjust processor-intensive operations more smoothly, without the wait for every frame to render at full quality causing jerkiness. You can set the auto proxy ratio by adjusting a slider in the General panel of the Fusion Preferences, found under the Fusion menu on macOS or the File menu on Windows and Linux.

Selective Updates

The last of the five buttons on the right of the transport controls is a three-way toggle that determines when nodes update images in the viewer. By default, when working in Fusion, any node needed to display the image in the viewer is updated. The Selective Update button can change this behavior during previews and final renders.

The three options are:

- **Update All (All):** Forces all the nodes in the current node tree to render. This is primarily used when you want to update all the thumbnails displayed in the Node Editor.
- **Selective (Some):** Causes only nodes that directly contribute to the current image to be rendered. So named because only selective nodes are rendered. This is the default setting.
- **No Update (None):** Prevents rendering altogether, which can be handy for making a lot of changes to a slow-to-render composition.

The options are also available in the Fusion Preferences General panel.

FUSION

Changing the Time Display Format

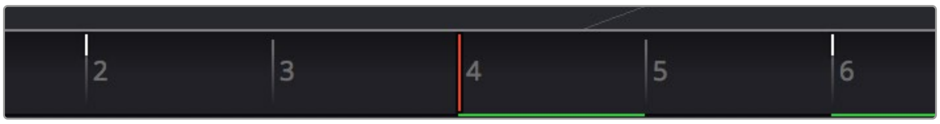
By default, all time fields and markers in Fusion count in frames, but you can also set the time display to SMPTE timecode or Feet + Frames.

To change the time display format:

1. Choose Fusion > Fusion Settings in DaVinci Resolve, choose Fusion Studio > Preferences in Fusion Studio on macOS, or choose File > Preferences in Fusion Studio on Windows or Linux.
2. When the Fusion settings dialog opens, select the Defaults panel and choose a Timecode option.
3. Select the Frame Format panel. If you're using timecode, choose a frame rate and turn on the "has fields" checkbox if your project is interlaced. If you're using feet and frames, set the Film Size value to match the number of frames found in a foot of film in the format used in your project.
4. Click Save.

Keyframe Display in the Time Ruler

When you select a node with keyframed parameters, those keyframes appear in the Time Ruler as little white tic marks, letting you navigate among and edit keyframes without having to open the Keyframes Editor or Spline Editor to see them.



The Time Ruler displaying keyframe marks

To move the playhead in the Time Ruler among keyframes:

- Press Option-Left Bracket ([) to jump to the next keyframe to the left.
- Press Option-Right Bracket (]) to jump to the next keyframe to the right.

The Fusion RAM Cache for Playback

When assembling a node tree, all image processing operations are rendered live to display the final result in the viewers. However, as each frame is rendered, and especially as you initiate playback forward or backward, these images are automatically stored to a RAM cache as they're processed so you can replay those frames in real time. The actual frame rate achieved during playback is displayed in the Status bar at the bottom of the Fusion window during playback. Of course, when you play beyond the cached area of the Time Ruler, uncached frames need to be rendered before being added to the cache.

Priority is given to caching nodes that are currently being displayed, based on which nodes are loaded to which viewers. However, other nodes may also be cached, depending on available memory and on how processor-intensive those nodes happen to be, among other factors.

Memory Limits of the RAM Cache

There is a single setting in DaVinci Resolve for limiting the RAM used for caching. This setting is located in the DaVinci Resolve Preferences Memory and GPU panel.

- **Limit Fusion Memory Cache To:** This slider sets the maximum amount of RAM that Fusion can access for caching. It is a subset of the RAM allocated to DaVinci Resolve. You can assign a maximum of 75% to Fusion from DaVinci Resolve's total RAM allocation. When not using the Fusion page, the RAM is released for other pages in DaVinci Resolve.

There are two settings in Fusion Studio for limiting the RAM used for caching. These settings are located in the Preferences Memory panel.

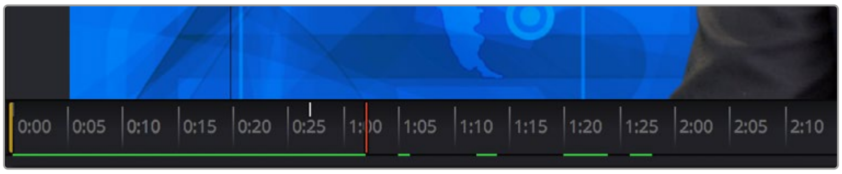
Limit Caching To: This slider sets the maximum amount of RAM used for caching. The 60% default setting on a 32-GB system limits the cache to 19.2 GB. The maximum amount you can assign to Fusion Studio is limited to 80% of the total system memory. This leaves a minimum amount of memory for other applications and the operating system.

Leave at least # MBytes: This number field further limits caching in cases where the system's available free RAM drops below the entered value. For instance, setting this to 200 MB attempts to keep 200 MB of RAM free for the OS or other applications. Setting the number field to 0 allows Fusion Studio to use the full amount of RAM specified by the Limit Caching To setting, ignoring other apps.

When the size of the cache reaches the Fusion Caching/Memory Limits setting found in the Memory panel of the Preferences, then lower-priority cache frames are automatically discarded to make room for new caching. You can keep track of the RAM cache usage via a percentage indicator on the far right of the Status bar at the bottom of the Fusion window.

Displaying Cached Frames

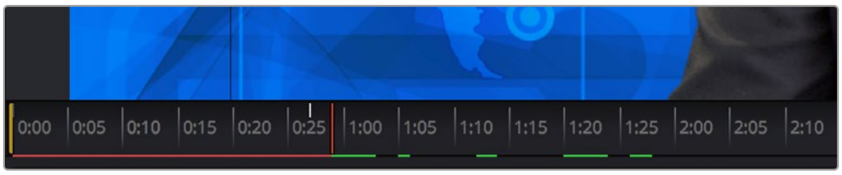
All cached frames for the currently viewed node are indicated by a green line at the bottom of the Time Ruler. Any green section of the Time Ruler should play back in real time.



The green lines indicate frames that have been cached for playback.

Temporarily Preserving the Cache When Changing Quality or Proxy Settings

If you toggle the composition's quality settings or proxy options, the cache is not immediately discarded. The green line instead turns red to let you know the cache is being preserved and can be used again when you go back to the original level of quality or disable proxy mode. However, if you play through those frames at the new quality or proxy settings, this preserved cache is overwritten with a new cache at the current quality or proxy setting.



A red line indicates that cached frames from a different quality or proxy setting are being preserved.

There's one exception to this, however. When you cache frames at the High Quality setting, and you then turn off High Quality, the green frames won't turn red. Instead, the High Quality cached frames are used even though the HiQ setting has been disabled.

FUSION

Toolbar

The toolbar, located underneath the Time Ruler, contains buttons that let you quickly add commonly used nodes to the Node Editor. Clicking any of these buttons adds that node after the currently selected node in the node tree, or adds an unconnected instance of that node if no nodes are selected. The Toolbar can be customized and saved for specific tasks.

The toolbar has buttons for adding commonly used nodes to the Node Editor.

The default toolbar is divided into sections that group commonly used nodes together. As you hover the pointer over any button, a tooltip shows you that node's name.

- **Loader/Saver nodes (Fusion Studio Only):** The Loader node is the primary node used to select and load clips from the hard drive. The Saver node is used to write or render your composition to disk.
- **Generator/Title/Paint nodes:** The Background and FastNoise generators are commonly used to create all kinds of effects, and the Title generator is obviously a ubiquitous tool, as is Paint.
- **Color/Blur nodes:** ColorCorrector, ColorCurves, HueCurves, and BrightnessContrast are the four most commonly used color adjustment nodes, while the Blur node is ubiquitous.
- **Compositing/Transform nodes:** The Merge node is the primary node used to composite one image against another. ChannelBooleans and MatteControl are both essential for reassigning channels from one node to another. Resize alters the resolution of the image, permanently altering the available resolution, while Transform applies pan/tilt/rotate/zoom effects in a resolution-independent fashion that traces back to the original resolution available to the source image.
- **Mask nodes:** Rectangle, Ellipse, Polygon, and BSpline mask nodes let you create shapes to use for rotoscoping, creating garbage masks, or other uses.
- **Particle system nodes:** Three particle nodes let you create complete particle systems when you click them from left to right. pEmitter emits particles in 3D space, while pMerge lets you merge multiple emitters and particle effects to create more complex systems. pRender renders a 2D result that can be composited against other 2D images.
- **3D nodes:** Seven 3D nodes let you build sophisticated 3D scenes. These nodes auto attach to one another to create a quick 3D template when you click from left to right. ImagePlane3D lets you connect 2D stills and movies for compositing into 3D scenes. Shape3D lets you create geometric primitives of different kinds. Text3D lets you build 3D text objects. Merge3D lets you composite multiple 3D image planes, primitive shapes, and 3D text together to create complex scenes, while Camera3D lets you frame the scene in whatever ways you like. SpotLight lets you light the scenes in different ways and Renderer3D renders the final scene and outputs 2D images and auxiliary channels that can be used to composite 3D output against other 2D layers.

When you're first learning to use Fusion, these nodes are really all you need to build most common composites. Once you've become a more advanced user, you'll still find that these are truly the most common operations you'll use.

Customizing the Toolbar

You can add and remove tools from the Fusion toolbar and then save the custom toolbar as a preset. New tools can be added by dragging them from the Effects Library or the Node Editor, and dividers can be added to group tool sets together.

To create a new toolbar, do the following:

1. Right click in an empty area of the toolbar and choose Customize > Create Toolbar from the contextual menu.
2. Enter a name for the toolbar in the dialog box, and click OK.

To rearrange the tools in the toolbar, do the following:

1. Create a new custom toolbar or select an existing custom toolbar.
2. Drag a node in the toolbar to a new location.

To add a tool to the toolbar:

1. Create a new custom toolbar or select an existing custom toolbar.
2. Do one of the following:
   - Drag a node from the Effects Library to the location on the toolbar where you want it added.
   - Drag a node from the Node Editor to the location on the toolbar where you want it added.

To add a divider to a toolbar, do the following:

1. Create a new custom toolbar or select an existing custom toolbar.
2. Right-click over any tool and choose Customize > Add Divider. A divider is added to the right of the tool.

To remove a tool from a toolbar, do the following:

1. Create a new custom toolbar or select an existing custom toolbar.
2. Right-click over any tool and choose Remove [name of tool].

To remove a group of tools between two dividers, do the following:

1. Create a new custom toolbar or select an existing custom toolbar.
2. Right-click over any tool in a group and choose Remove Group.

To prevent a custom toolbar from being modified:

Right-click over the toolbar and choose Lock from the contextual menu.

To switch between toolbars:

Right-click over the toolbar and choose the custom toolbar name or choose Default to return to Fusion's default toolbar.

To rename a custom toolbar, do the following:

1. Right-click over the toolbar and choose the name of the custom toolbar you want to rename.
2. Right-click over the toolbar again and choose Customize > Rename [toolbar name].
3. Enter a new name for the toolbar.

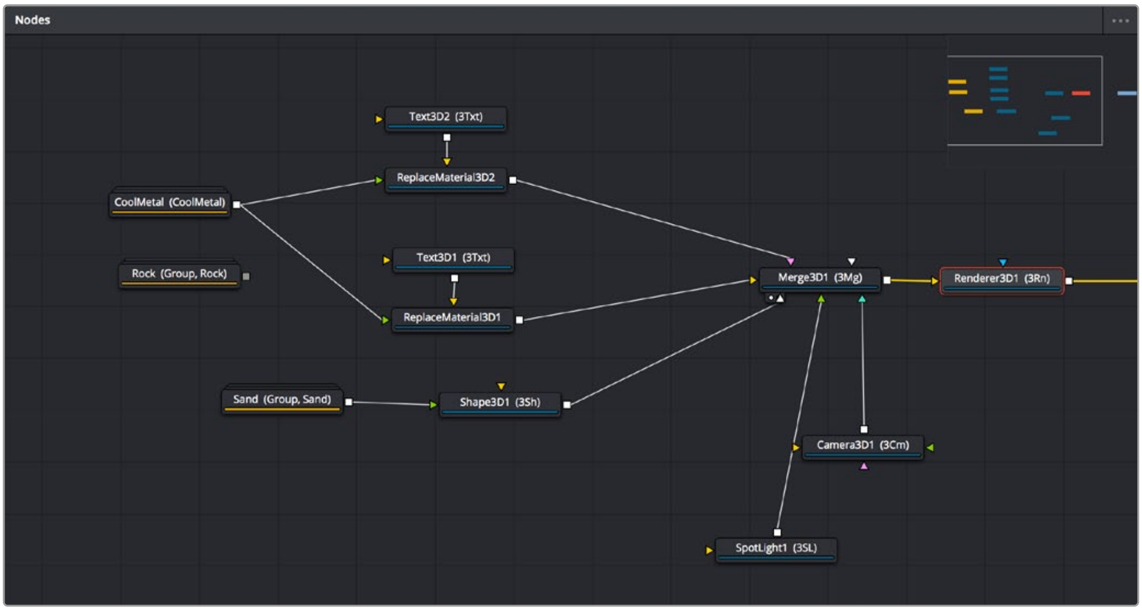
To delete a custom toolbar, do the following:

1. Right-click over the toolbar and choose the name of the custom toolbar you want to delete.
2. Right-click over the toolbar again and choose Customize > Remove [toolbar name].

> **TIP:** Adding and deleting tools from a custom toolbar is not undoable. If you are creating a complex toolset, make a new custom toolbar based on your current toolbar in between major changes and work off that. That way if you make an error, you can revert back to the last known good toolbar. Once you have the final toolbar the way you want it, you can go back and remove all the interim custom toolbars you made.

Node Editor

The Node Editor is the heart of Fusion because it's where you build the tree of nodes that makes up each composition. Each node you add to the node tree adds a specific operation that creates one effect, whether it's blurring the image, adjusting color, painting strokes, drawing and adding a mask, extracting a key, creating text, or compositing two images into one.



The Node Editor displaying a node tree creating a composition

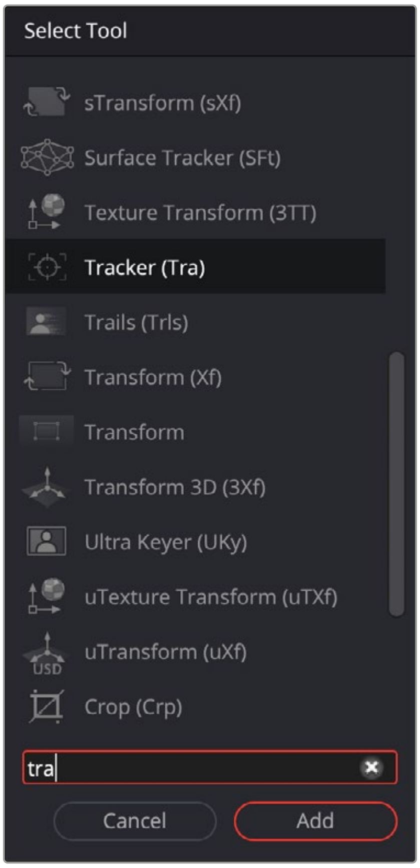
You can think of each node as a layer in an effects stack, except that you have the freedom to route image data in any direction to branch and merge different segments of your composite in completely nonlinear ways. This makes it easy to build complex effects, but it also makes it easy to see what's happening, since the node tree doubles as a flowchart that clearly shows you everything that's happening, once you learn to read it.

Adding Nodes to Your Composition

Depending on your mood, there are a few ways you can add nodes from the Effects Library to your composition. For most of these methods, if there's a single selected node in the Node Editor, new nodes are automatically added after it, but if there are no selected nodes or multiple selected nodes, then new nodes are added as disconnected from anything else.

Methods of adding nodes include:

- Click a button in the toolbar.
- Open the Effects Library, find the node you want in the relevant category, and click once on a node you'd like to add.
- Right-click on a node and choose Insert Tool from the drop-down menu to add it after the node you've right-clicked on. Or, you can right-click on the background of the Node Editor to use that submenu to add a disconnected node.
- Press Shift-Spacebar to open a Select Tool dialog, type characters corresponding to the name of the node you're looking for, and press the Return key (or click OK) when it's found. Once you learn this method, it'll probably become one of your most frequently used ways of adding nodes.



The Select Tool dialog lets you find any node quickly if you know its name.

Removing Nodes from Your Composition

Removing nodes is as simple as selecting one or more nodes, and then pressing the Delete or Backspace keys.

Identifying Node Inputs and Node Outputs

Each node displays small colored connections around the edges. One or more arrows represent inputs, and the square connection represents the tool's processed output, of which there is always only one. If you hover the pointer over any of a node's inputs or output, the name of that input or output immediately appears in the Status bar. If you wait for a few more moments, a floating tooltip displays the same name right over the node.

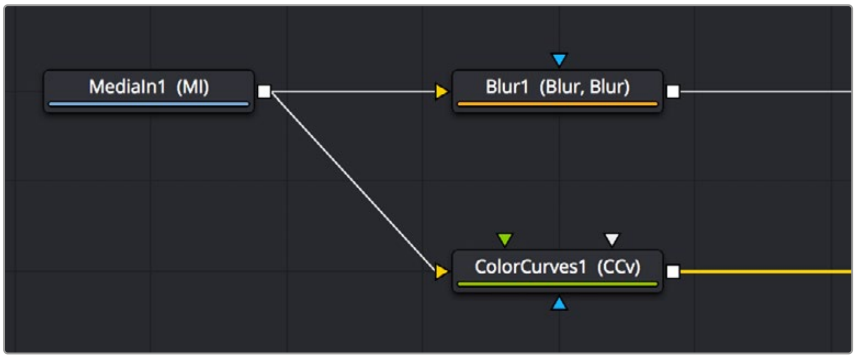
Node Editing Essentials

Each node has inputs and outputs that are "wired together" using connections. The inputs are represented by arrows that indicate the flow of image data from one node to the next, as each node applies its effect and feeds the result (via the square output) to the next node in the tree. In this way, you can quickly build complex results from a series of relatively simple operations.

Three nodes connected together

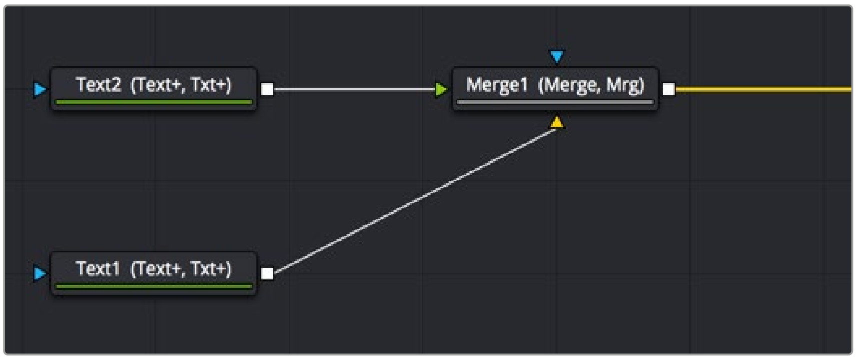
FUSION

You can connect a single node's output to the inputs of multiple nodes (called "branching").



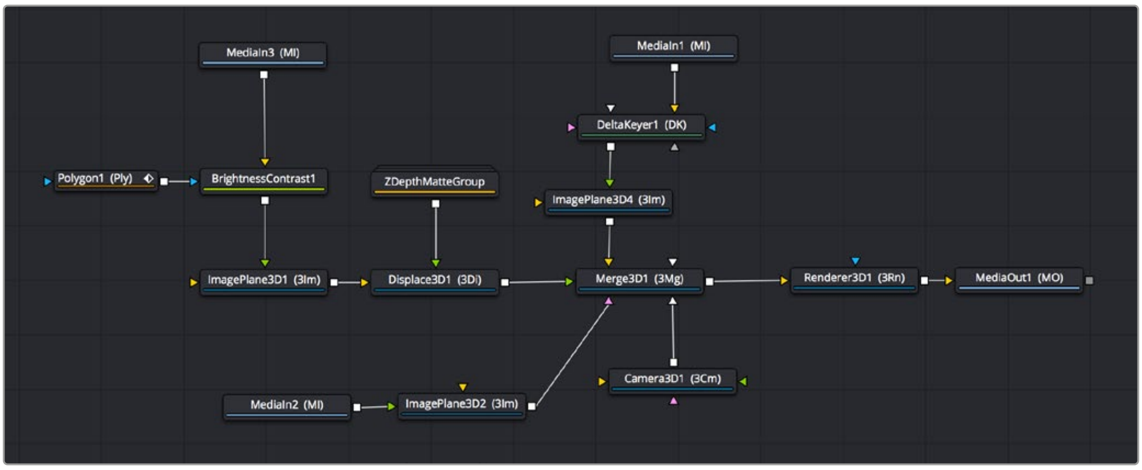
One node branching to two to split the image to two operations

You can then composite images together by connecting the output from multiple nodes to certain nodes such as the Merge node that combines multiple inputs into a single output.



Two nodes being merged together into one to create a composite

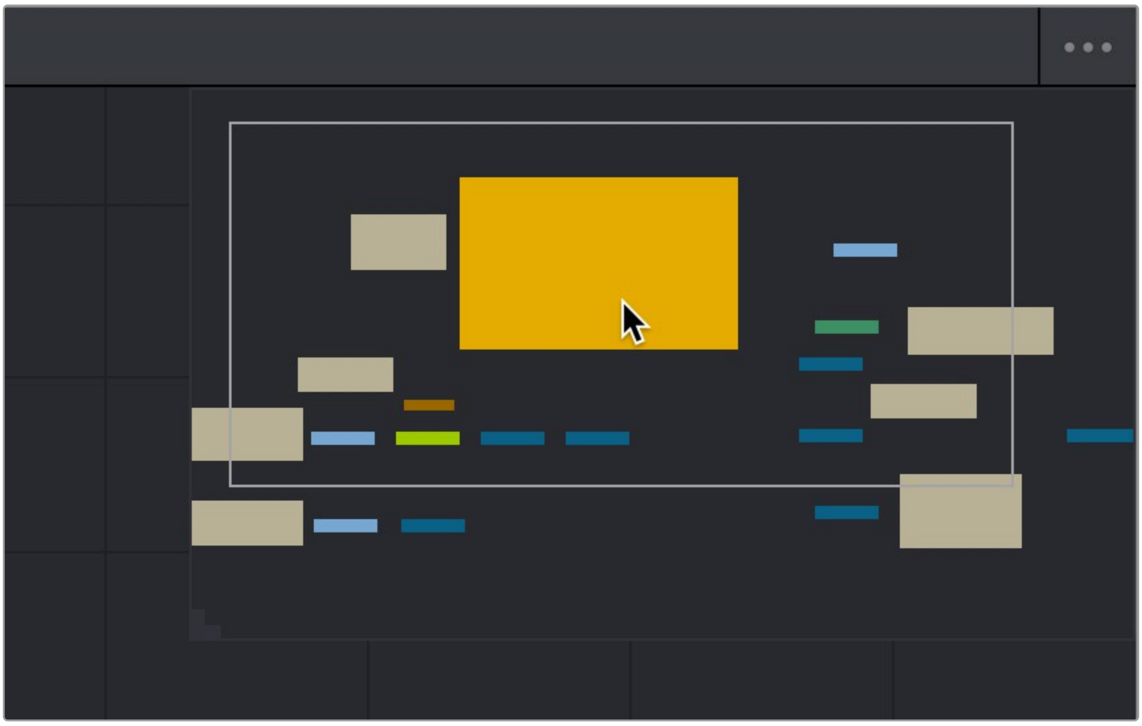
By default, new nodes are added from left to right in the Node Editor, but they can also flow from top to bottom, right to left, bottom to top, or in all directions simultaneously. Connections automatically reorient themselves along all four sides of each node to maintain the cleanest possible presentation as you rearrange other connected nodes.



Nodes can be oriented in any direction; the input arrows let you follow the flow of image data.

Navigating the Node Editor

As your Node tree gets larger, parts inevitably go offscreen. When a portion of the node tree is offscreen, a resizable Navigator pane appears in the upper-right corner. The Navigator is a miniature representation of the entire node tree that you can drag within to pan to different parts of your composition quickly. You can resize the navigator using a handle in the lower-left corner, and you can choose to show or hide the Navigator by pressing the V key, or by right-clicking the Node Editor to access the Options submenu of the contextual menu.



The Navigator pane for accessing offscreen parameters or tools

There are other standard methods of panning and zooming around the Node Editor.

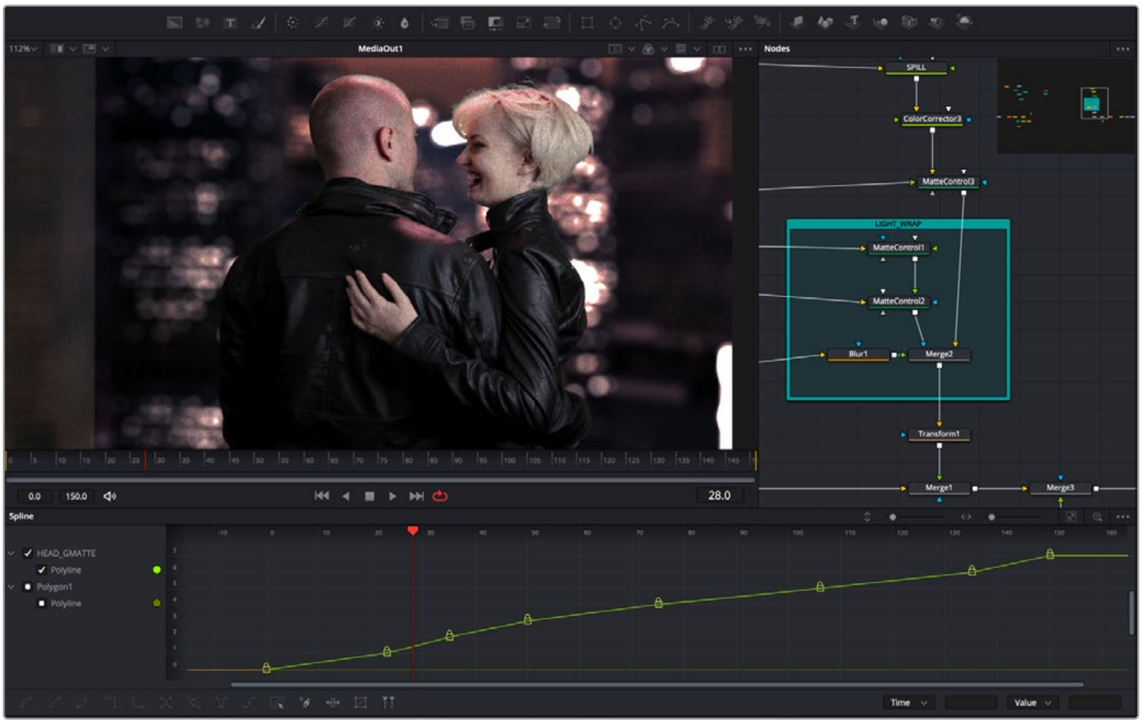
Methods of navigating the Node Editor:

- Middle-click and drag to pan around the Node Editor.
- Hold down Shift and Command and drag the Node Editor to pan.
- Press the Middle and Left mouse buttons simultaneously and drag to resize the Node Editor.
- Hold down the Command key, and use your mouse's scroll control to resize the Node Editor.
- Right-click the Node Editor and choose an option from the Scale submenu of the contextual menu.
- Press Command-1 to reset the Node Editor to its default size.
- Drag two fingers on a track pad to pan.
- Hold the Command key down and drag two fingers on a track pad to resize the Node Editor.

Vertical Node Editor Layouts

Alternative Node View Layout presets located in the Fusion page allow for positioning the Node Editor vertically, either alongside the Inspector, or along the left side of the screen. This can be very helpful when animating in the Spline Editor or Keyframes Editor.

When in the Fusion page, you can choose the layouts from Workspace > Layout Presets. Choosing a vertical layout allows the node tree to flow from top to bottom, leaving much more room along the lower half of the screen for the Spline Editor or Keyframes Editor.



The Mid Flow Vertical layout preset used with the Vertical Flow direction setting.

When using the vertical layouts, enabling the Flow > Build Direction > Vertical option in the Fusion settings will cause all new Node trees to build vertically, leaving maximum room for Fusion's animation tools.

You can then save alternative layouts based on these two vertical presets using the Workspace > Layout Presets submenu.

When you want to return to the default horizontal Node Editor layout, just choose Workspace > Layout Presets > Fusion Presets > Default.

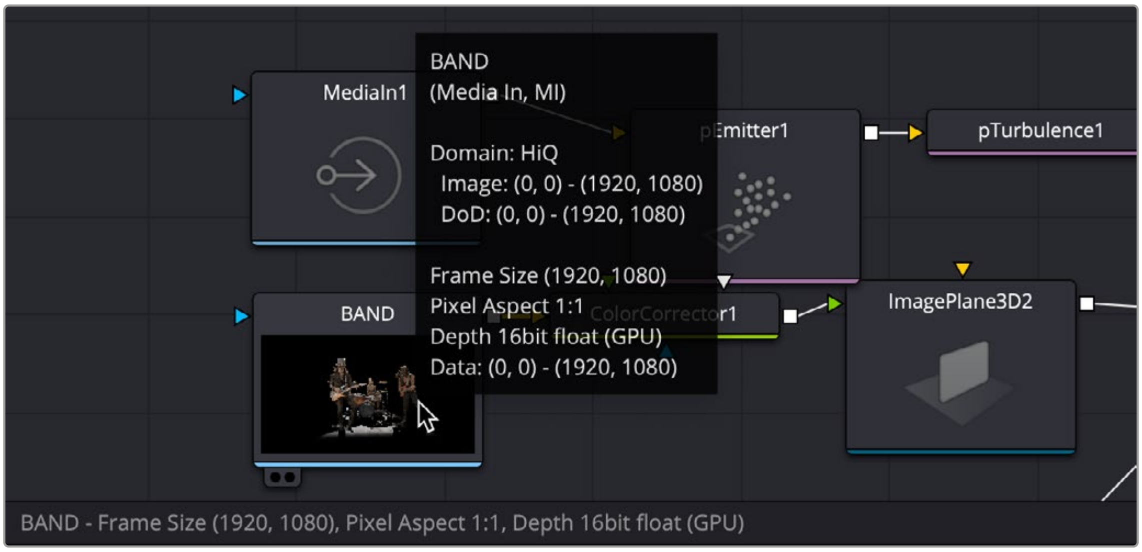
These Layout options are not available in Fusion Studio, however, you can use the Floating Frame to position the Node Editor wherever you like.

Keeping Organized

As you work, it's important to keep the node trees that you create tidy to facilitate a clear understanding of what's happening. Fortunately, the Fusion Node Editor provides a variety of methods and options to help you with this, found within the Options and Arrange Tools submenus of the Node Editor contextual menu.

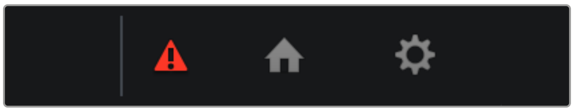
Status Bar

The Status bar in the lower-left corner of the Fusion window shows you a variety of up-to-date information about things you're selecting and what's happening in Fusion. For example, hovering the pointer over a node displays information about that node in the Status bar. Additionally, the currently achieved frame rate appears whenever you initiate playback, and the percentage of the RAM cache that's used appears at all times. Other information, updates, and warnings appear in this area as you work.



The Status bar under the Node Editor showing you information about a node under the pointer, as well as a popup box with even more relevant info.

Occasionally the Status bar will display a badge to let you know there's a message in the console you might be interested in. The message could be a log, script message, or error.

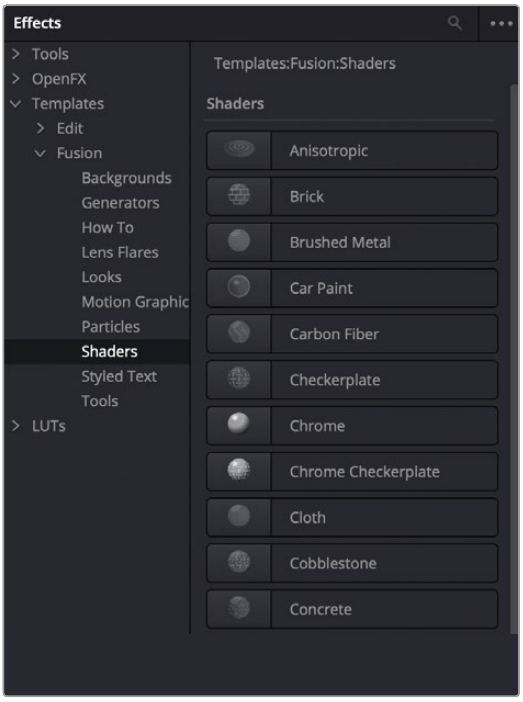


A notification that there's a message in the Console

Effects Library

The Effects Library in Fusion shows all the nodes and effects available in Fusion, including third-party OFX plugins, if installed. If you are using DaVinci Resolve, Resolve FX also appear in the OFX category. While the toolbar shows many of the most common nodes you'll use in any composite, the Effects Library contains every single tool available in Fusion, organized by category, with each node ready to be quickly added to the Node Editor. Suffice it to say that there are many, many more nodes available in the Effects Library than on the toolbar, spanning a wide range of uses.

The Effects Library with Tools open



The Templates section of the Effects Library

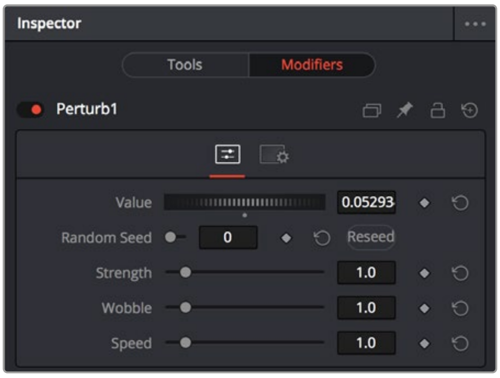
The hierarchical category browser of the Effects Library is divided into several sections depending on whether you are using Fusion Studio or the Fusion page within DaVinci Resolve. The Tools section is the most often used since it contains every node that represents an elemental image-processing operation in Fusion. Hovering the pointer over a specific tool will reveal a tool-tip explaining its functionality at the bottom right of the DaVinci Resolve interface. The OpenFX section contains third-party plugins, and if you are using the Fusion page, it also contains ResolveFX, which are included with DaVinci Resolve. A third section, only visible when using the Fusion page in DaVinci Resolve, is the Templates section. The Template section contains a variety of additional content including templates for Lens Flares, Backgrounds, Generators, Particle Systems, Shaders (for texturing 3D objects), and other resources for use in your composites.

> The Effects Library's list can be made full height or half height using a button at the far left of the UI toolbar.

The Inspector

The Inspector is a panel on the right side of the Fusion window that you use to display and manipulate the parameters of one or more selected nodes. When a node is selected in the Node Editor, its parameters and settings appear in the Inspector.

The Inspector shows parameters from one or more selected nodes.



The Modifier panel showing a Perturb modifier

The Tools and Modifiers Panels

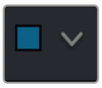
The Fusion Inspector is divided into two panels. The Tools panel is the main panel that shows you the parameters of selected nodes. The Modifiers panel shows optional extensions to the standard parameters of a tool. In the following image, a Perturb modifier has been added to a parameter to add random animation to that parameter, and the controls found on the Modifier panel let you customize what kind of randomness is being added.

> Other nodes display node-specific items here. For example, Paint nodes show each brush stroke as an individual set of controls in the Modifiers panel, available for further editing or animating.

Parameter Header Controls

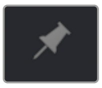
A cluster of controls appears at the top of every node's controls in the Inspector.

Common Inspector controls

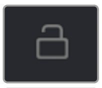


Set Color: A pop-up menu that lets you assign one of 16 colors to a node, overriding a node's own color.

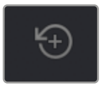
Versions: Clicking Versions reveals another toolbar with six buttons. Each button can hold an individual set of adjustments for that node that you can use to store multiple versions of an effect.



Pin: The Inspector is also capable of simultaneously displaying all parameters for multiple nodes you've selected in the Node Editor. Furthermore, a Pin button in the title bar of each node's parameters lets you "pin" that node's parameters into the Inspector so that they remain there even when that node is deselected, which is valuable for key nodes that you need to adjust even while inspecting other nodes of your composition.



Lock: Locks that node so that no changes can be made to it.



Reset: Resets all parameters within that node.

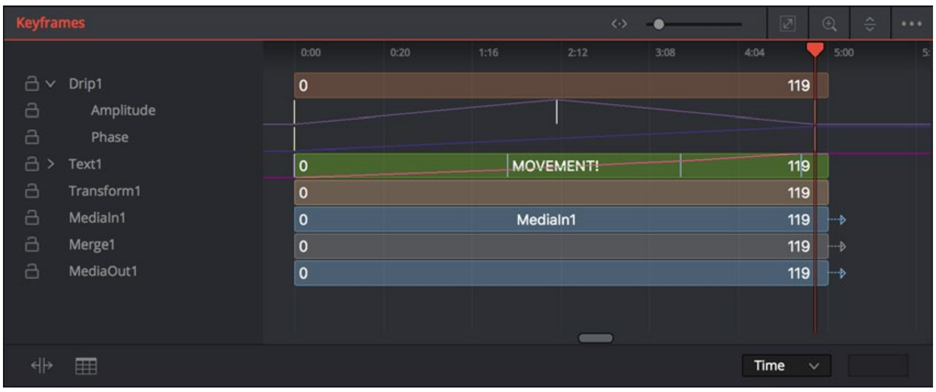
Parameter Tabs

Many nodes expose multiple tabs' worth of controls in the Inspector, seen as icons at the top of the parameter section for each node. Click any tab to expose that set of controls.

Nodes with several tabs' worth of parameters

Keyframes Editor

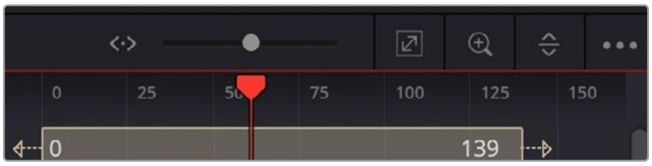
The Keyframes Editor displays each node in the current composition as a stack of layers within a miniature timeline. The order of the layers is largely irrelevant as the order and flow of connections in the node tree dictate the order of image-processing operations. You use the Keyframes Editor to trim, extend, or slide Loader, MediaIn, and effects nodes, or to adjust the timing of keyframes, which appear superimposed over each effect node unless you open them up into their editable track.



The Keyframes Editor is used to adjust the timing of clips, effects, and keyframes.

Keyframes Editor Control Summary

At the top, a series of zoom and framing controls let you adjust the work area containing the layers.



These controls let you manipulate the Keyframes Editor view.

A Horizontal zoom control lets you scale the size of the editor.

- A Zoom to Fit button fits the width of all layers to the current width of the Keyframes Editor.
- A Zoom to Rect tool lets you draw a rectangle to define an area of the Keyframes Editor to zoom into.
- A Sort pop-up menu lets you sort or filter the tracks in various ways.
- An Option menu provides access to many other ways of filtering tracks and controlling visible options.

A timeline ruler provides a time reference, as well as a place in which you can scrub the playhead.

At the left, a track header contains the name of each layer, as well as controls governing that layer.

- A lock button lets you prevent a particular layer from being changed.
- Nodes that have been keyframed have a disclosure control, which when opened displays a keyframe track for each animated parameter.

In the middle, the actual editing area displays all layers and keyframe tracks available in the current composition.

At the bottom-left, Time Stretch and Spreadsheet mode controls provide additional ways to manipulate keyframes.

At the bottom right, the Time/TOffset/TScale drop-down menu and value fields let you numerically alter the position of selected keyframes either absolutely, relatively, or based on their distance from the playhead.

Adjusting Clip Timings

Each Loader or MediaIn node that represents a clip used in a composition is represented as a layer in this miniature timeline. You can edit a layer's In or Out points by positioning the pointer over the beginning or end of a segment and using the resize cursor to drag that point to a new location. You can slide a layer by dragging it to the left or right, to better line up with the timing of other elements in your composition.

The Keyframes Editor also lets you adjust the timing of elements that you've added from directly within Fusion.

Adjusting Effect Timings

Each effect node also appears as a layer, just like clips. You can resize the In and Out points of an effect layer, and slide the entire layer forward or backward in time, just like a Loader or MediaIn layers. If you trim an effect layer to be shorter than the duration of the composition, the effect cuts in at whichever frame the layer begins, and cuts out after the last frame of that layer, just like a clip on a timeline.

Adjusting Keyframe Timings

When you've animated an effect by adding keyframes to a parameter in the Inspector, the Keyframes Editor is used to edit the timing of keyframes easily. By default, all keyframes applied to parameters within a particular node's layer appear superimposed in one flat track over the top of that layer.

To edit keyframes, you can click the disclosure control to the left of any animated layer's name in the track header, which opens up keyframe tracks for every keyframed parameter within that layer.

Keyframe tracks exposed

Keyframe Editing Essentials

Here's a short list of keyframe editing methods that will get you started.

Methods of adjusting keyframes:

- You can click on a single keyframe to select it.
- You can drag a bounding box over a series of keyframes to select them all.
- You can drag keyframes left and right to reposition them in time.
- You can right-click one or more selected keyframes and use commands from the drop-down menu to change keyframe interpolation, copy/paste keyframes, or even create new keyframes.
- You can Command-drag one or more selected keyframes to drag a duplicate of them to another position in the keyframe track.

To change the position of a keyframe using the toolbar, do one of the following:

- Select a keyframe, and then enter a new frame number in the Time Edit box.
- Choose T Offset from the Time Editor pop-up, select one or more keyframes, and enter a frame offset.
- Choose T Scale from the Time Editor pop-up, select one or more keyframes, and enter a multiplier added to the current playhead frame position. For instance, if the playhead is on frame 10 and the keyframe is on frame 30, entering the TScale value of 2 will position the keyframe on frame 50. The distance between the playhead and original keyframe is 20, so (20 x 2) = 40, which is then added to the playhead position.

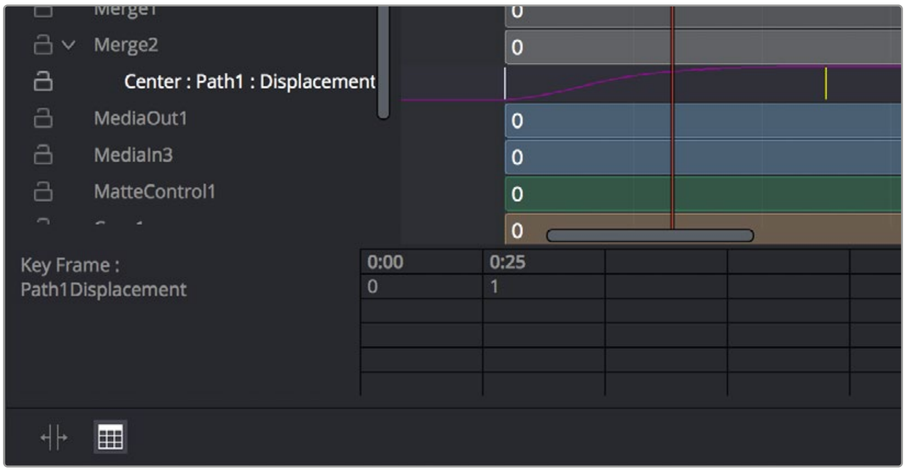
Time Stretching Keyframes

If you select a range of keyframes in a keyframe track, you can turn on the Time Stretch tool to show a box used to squeeze and stretch the entire range of keyframes relative to one another. The Time Stretcher changes the overall timing of a sequence of keyframes without losing the relative timing from one keyframe to the next. Alternatively, you can turn on Time Stretch and draw a bounding box around the keyframes you want to adjust to create a time-stretching boundary that way. Click the Time Stretch tool again to turn it off.

Time stretching keyframes

The Keyframe Spreadsheet

If you turn on the Spreadsheet and then click on the name of a layer in the keyframe track, the numeric time position and value (or values if it's a multi-dimensional parameter) of each keyframe appear as entries in the cells of the Spreadsheet. Each column represents one keyframe, while each row represents a single aspect of each keyframe.



Editing keyframes in the Spreadsheet

For example, if you're animating a motion path, then the "Key Frame" row shows the frame each keyframe is positioned at, and the "Path1Displacement" row shows the position along the path at each keyframe. If you change the Key Frame value of any keyframe, you'll move that keyframe to a new frame of the Timeline.

Spline Editor

The Spline Editor provides a more detailed environment for editing the timing, value, and interpolation of keyframes. Using control points at each keyframe connected by splines (also called curves), you can adjust how animated values change over time. The Spline Editor has four main areas: the Zoom and Framing controls at the top, the Parameter list at the left, the Graph Editor in the middle, and the toolbar at the bottom.

The Spline Editor is divided into the Zoom controls at top, Parameter list at left, Graph Editor, and toolbar.

Spline Editor Control Summary

At the top, a series of Zoom and Framing controls let you adjust the work area containing the curves.

- Vertical and horizontal zoom controls let you scale the size of the editor.
- A Zoom to Fit button fits the width of all curves to the current width of the Spline Editor.
- A Zoom to Rect tool lets you draw a rectangle to define an area of the Spline Editor to zoom into.

A Timeline ruler provides a time reference, as well as a place in which you can scrub the playhead.

The Parameter list at the left is where you decide which splines are visible in the Graph view. By default, the Parameter list shows every parameter of every node in a hierarchical list. Checkboxes beside each name are used to show or hide the curves for different keyframed parameters. Color controls let you customize each spline's tint to make splines easier to see in a crowded situation.

The Graph view that takes up most of this panel shows the animation spline along two axes. The horizontal axis represents time, and the vertical axis represents the spline's value. Selected control points show their values in the edit fields at the bottom of the graph.

Lastly, the toolbar at the bottom of the Spline Editor provides controls to set control point interpolation, spline looping, or choose Spline editing tools for different purposes.

Choosing Which Parameters to Show

Before you start editing splines to customize or create animation, you need to choose which parameter's splines you want to work on.

To show every parameter in every node:

- Click the Splines Editor Option menu and choose Expose All Controls. Toggle this control off again to go back to viewing what you were looking at before.

To show splines for the currently selected node:

- Click the Splines Editor Option menu and choose Show Only Selected Tool.

Essential Spline Editing

The Spline Editor is a deep and sophisticated environment for keyframe and spline editing and retiming, but the following overview will get you started using this tool for creating and refining animation.

To select one or more control points:

- Click any control point to select it.
- Command-click multiple control points to select them.
- Drag a bounding box around multiple control points to select them as a group.

To edit control points and splines:

- Click anywhere on a spline to add a control point.
- Drag one or more selected control points to reshape the spline.
- Shift-drag a control point to constrain its motion vertically or horizontally.

To edit Bézier curves:

- Select any control point to make its Bézier handles visible, and drag the Bézier handles.
- Command-drag a Bézier handle to break the angle between the left and right handles.

To delete control points:

- Select one or more control points and press the Delete or Backspace key.

Essential Spline Editing Tools and Modes

The Spline Editor toolbar at the bottom contains a mix of control point interpolation buttons, Spline loop modes, and Spline editing tools.

Control Point Interpolation

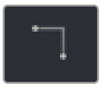
The first six buttons let you adjust the interpolation of one or more selected control points.

Control point interpolation controls

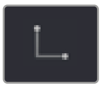
Smooth: Creates automatically adjusted Bézier curves to create smoothly interpolating animation.

Linear: Creates linear interpolation between control points.

Invert: Inverts the vertical position of non-animated LUT splines. This does not operate on animation splines.



Step In: For each keyframe, creates sudden changes in value at the next keyframe to the right. Similar to a hold keyframe in After Effects® or a static keyframe in the DaVinci Resolve Color page.



Step Out: Creates sudden changes in value at every keyframe for which there's a change in value at the next keyframe to the right. Similar to a hold keyframe in After Effects or a static keyframe in the DaVinci Resolve Color page.

Reverse: Reverses the horizontal position of selected keyframes in time, so the keyframes are backward.

Spline Loop Modes

The next three buttons let you set up spline looping after the last control point on a parameter's spline, enabling a limited pattern of keyframes to animate over a far longer duration. Only the control points you've selected are looped.

Spline Loop modes

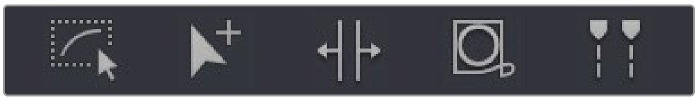
Set Loop: Repeats the same pattern of keyframes over and over.

Set Ping Pong: Repeats a reversed set of the selected keyframes and then a duplicate set of the selected keyframes to create a more seamless pattern of animation.

Set Relative: Repeats the same pattern of selected keyframes but with the values of each repeated pattern of keyframes being incremented or decremented by the trend of all keyframes in the selection. This results in a loop of keyframes where the value either steadily increases or decreases with each subsequent loop.

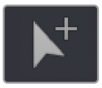
Spline Editing Tools

The next five buttons provide specialized Spline editing tools.

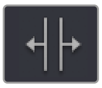


Spline editing controls

Select All: Selects every keyframe currently available in the Spline Editor.

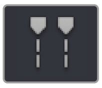


Click Append: Click once to select this tool and click again to de-select it. This tool lets you add or adjust keyframes and spline segments (sections of splines between two keyframes), depending on the keyframe mode you're in. With Smooth or Linear keyframes, clicking anywhere above or below a spline segment adds a new keyframe to the segment at the location where you clicked. With Step In or Step Out keyframes, clicking anywhere above or below a line segment moves that segment to where you've clicked.



Time Stretch: If you select a range of keyframes, you can turn on the Time Stretch tool to show a box you can use to squeeze and stretch the entire range of keyframes relative to one another, to change the overall timing of a sequence of keyframes without losing the relative timing from one keyframe to the next. Alternatively, you can turn on Time Stretch and draw a bounding box around the keyframes you want to adjust to create a time-stretching boundary that way. Click Time Stretch a second time to turn it off.

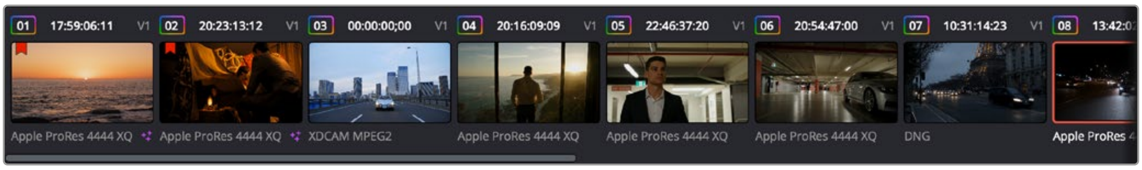
Shape Box: Turn on the Shape Box to draw a bounding box around a group of control points you want to adjust in order to horizontally squish and stretch (using the top/bottom/left/right handles), corner pin (using the corner handles), move (dragging on the box boundary), or corner stretch (Command-drag the corner handles).



Show Key Markers: Turning on this control shows keyframes in the top ruler that correspond to the frame at which each visible control point appears. The colors of these keyframes correspond to the color of the control points they're indicating.

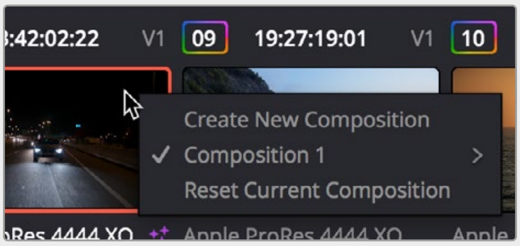
Thumbnail Timeline in the Fusion Page

In the Fusion page of DaVinci Resolve, the Thumbnail timeline (hidden by default) can be opened by clicking the Clips button in the UI toolbar and appears underneath the Node Editor when it's open. The Thumbnail timeline shows every clip in the current Timeline, giving you a way to navigate from one clip to another. Each thumbnail has a pop-up menu for creating and switching among multiple versions of compositions, and resetting the current composition, when necessary.



The Thumbnail timeline lets you navigate the Timeline and manage versions of compositions.

Right-clicking on any thumbnail exposes a contextual menu.



The contextual menu for the Thumbnail timeline

To open another clip:

- Click any thumbnail to jump to that clip's composition. The current clip is outlined in orange.

To navigate the Thumbnail timeline:

- Middle-click and drag the Timeline left or right to scroll through the clips in the Timeline.
- Click and drag the slider at the bottom of the Thumbnail timeline to scroll through the clips in the Timeline.

To create and manage versions of compositions:

- **To create a new version of a composition:** Right-click the current thumbnail, and choose Create New Composition from the contextual menu.
- **To load a different composition:** Right-click the current thumbnail, and choose "NameOfVersion" > Load from the contextual menu.
- **To delete a composition:** Right-click the current thumbnail, and choose "NameOfVersion" > Delete from the contextual menu.
- **To rename a composition:** Right-click the current thumbnail, and choose "NameOfVersion" > Rename from the contextual menu.

To reset the current composition:

- Right-click the current thumbnail, and choose Reset Current Composition from the contextual menu.

To change how thumbnails are identified:

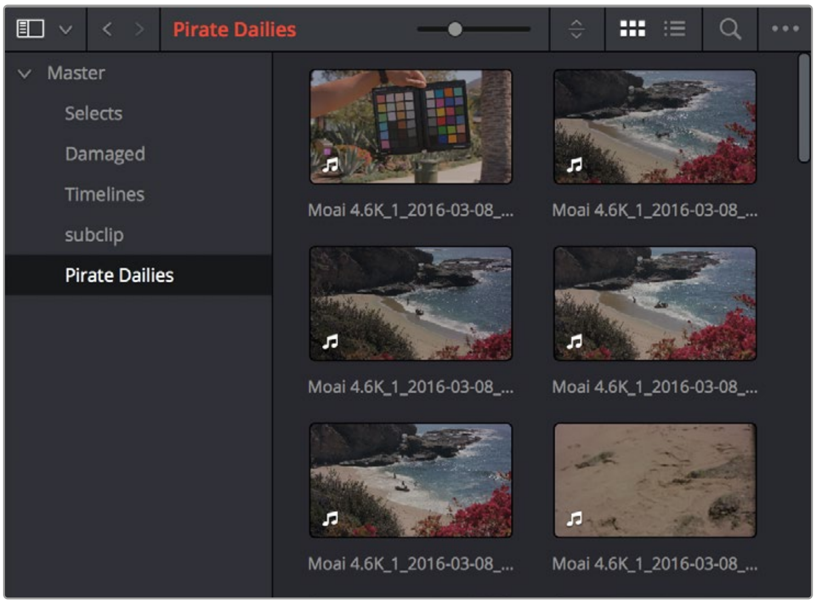
- Double-click the area underneath any thumbnail to toggle among clip format, clip name, and the composition version name.

The Media Pool in the Fusion Page

In DaVinci Resolve's Fusion page, the Media Pool continues to serve its purpose as the repository of all media you've imported into your project. This makes it easy to add additional clips to your compositions simply by dragging the clip you want from the Media Pool into the Node Editor. The media you add appears as a new MediaIn node in your composition, ready to be integrated into your node tree however you need it.

TIP: If you drag one or more clips from the Media Pool onto a connection line between two nodes in the Node Editor, the clips are automatically connected to that line via enough Merge nodes to connect them all.

For more information on using the myriad features of the Media Pool, *see Chapter 18, "Adding and Organizing Media with the Media Pool."*



The Media Pool in Thumbnail mode showing video clips

Importing Media Into the Media Pool on the Fusion Page

If you find yourself in the Fusion page and you need to quickly import a few clips for immediate use, you can do so in a couple of different ways.

To add media by dragging one or more clips from the Finder to the Fusion page Media Pool (macOS only):

1 Select one or more clips in the Finder.

2 Drag those clips into the Media Pool of DaVinci Resolve, or to a bin in the Bin list. Those clips are added to the Media Pool of your project.

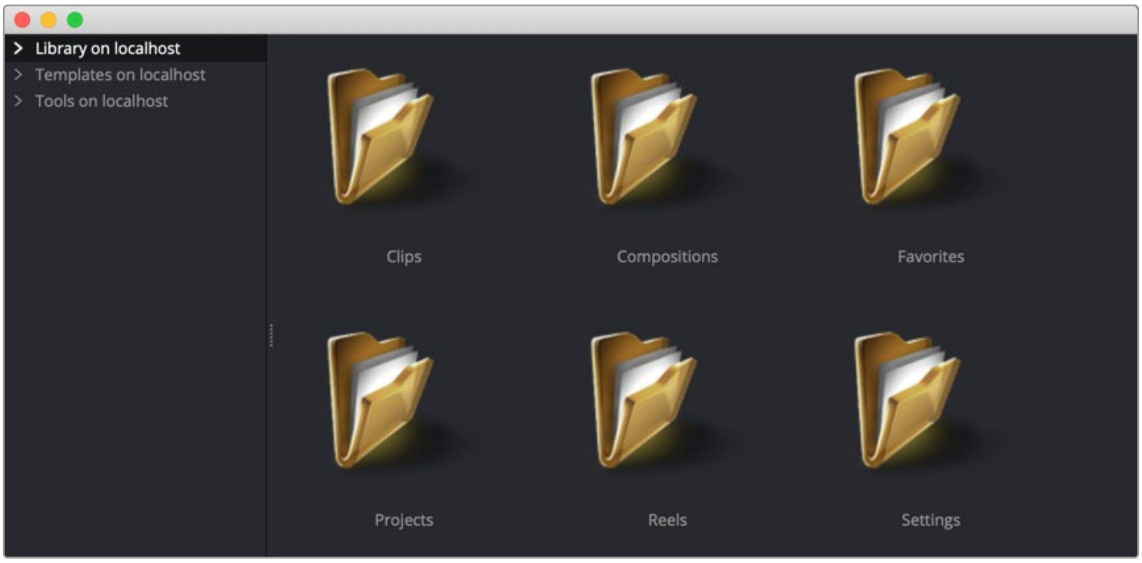
To use the Import Media command in the Fusion page Media Pool:

1. With the Fusion page open, right-click anywhere in the Media Pool, and choose Import Media.
2. Use the Import dialog to select one or more clips to import, and click Open. Those clips are added to the Media Pool of your project.

For more information on importing media using the myriad features of the Media page, *see Chapter 18, "Adding and Organizing Media with the Media Pool."*

Bins in Fusion Studio

Bins in Fusion Studio are similar to the Media Pool in DaVinci Resolve. Bins are organizational panels that provide an easy way of accessing commonly used tools, settings, macros, compositions, and media content. They can keep all your custom content and resources close at hand, so you can use them without searching through your hard drives. Bins can also be shared over a network to improve a collaborative workflow with other Fusion Studio artists.



The Bins window

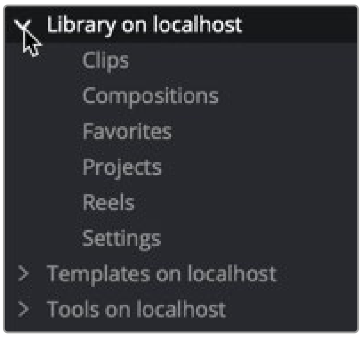
To open the Bins window:

Choose File > Bins from the menu bar.

Similar to the Media Pool in DaVinci Resolve, when adding an item to the Fusion bins, a link is created between the item on disk and the bins. Fusion does not copy the file into its own cache or hard drive space. The file remains in its original format and in its original location.

Bins Interface

The Bins window is actually a separate application used to save content you may want to reuse at a later time. The Bins window is separated into two panels. The sidebar on the left is a bin list where items are placed into categories, while the panel on the right displays the selected bin's content.



The Bin list sidebar

The Bin list organizes content into bins or folders using a hierarchical List view. These folders can be organized to suit your workflow, but standard folders are provided for Clips, Compositions, Favorites, Settings, and Tools. Parent folders contain subfolders that hold the content. For instance, the Tools bin is a parent folder to all the categories of tools. To access subfolders, click the disclosure arrow to the left of the parent folder's name.

When you select a bin from the bin list, the contents of the folder are displayed in the Contents panel as thumbnail icons.

The Bins icon view

A toolbar along the bottom of the bin provides access to organization, playback, and editing controls.

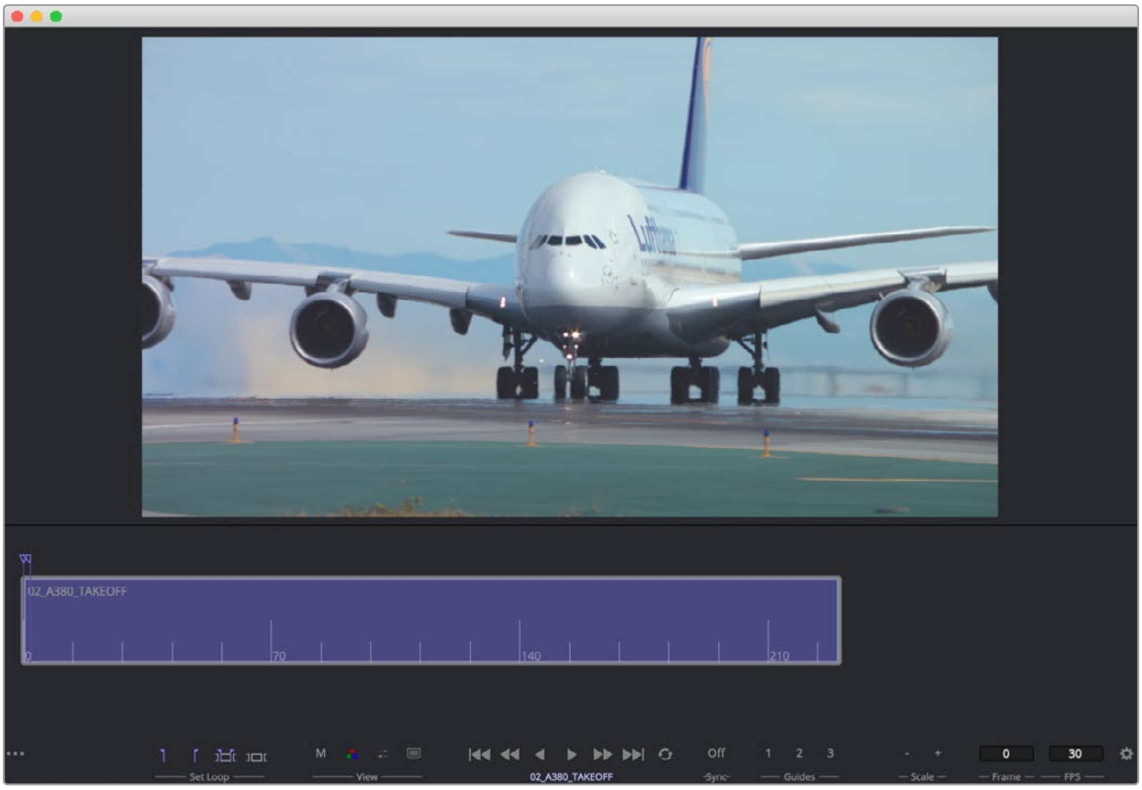
The Bins toolbar

- **New Folder:** Creates a new folder in the current window.
- **New Reel:** Creates an empty reel that can contain multiple clips edited together into a timeline.
- **New Clip:** Opens a dialog to link a new media file into a bin.
- **Studio Player:** Opens a playback viewer for a selected clip.
- **Icon/List view:** This button toggles between showing contents of a bin in thumbnail view and list view.
- **Checkerboard:** Shows a checkerboard pattern in a clip thumbnail to signify transparency.
- **Thumbnail size:** Provides a few preset sizes for thumbnail icons.

Right-clicking in any area of the bin window displays a pop-up menu to access most of a bin's features. Right-clicking on an item in a bin shows the same menu with additional options for renaming, playing, or deleting the item.

The Bin Studio Player

Selecting a clip in a bin and clicking the Studio Player button or double-clicking the clip opens the Studio Player. The Studio Player can be used to view clips, view metadata, and add notes.



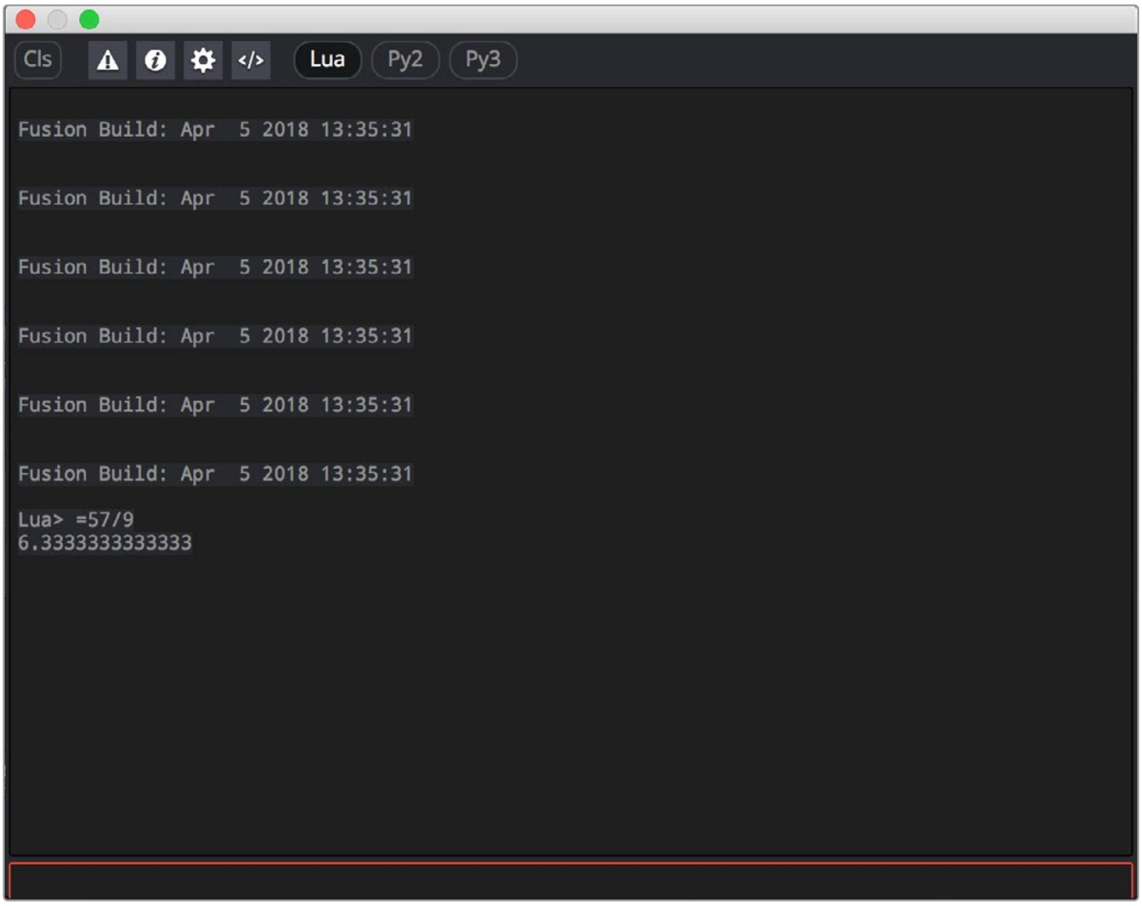
The Bin Studio Player

For more information on Bins, see Chapter 13, "Bins" in the *Fusion Referance Manual*.

The Console

The Console is a window in which you can see the error, log, script, and input messages that may explain something Fusion is trying to do in greater detail. The Console is also where you can read FusionScript outputs, or input FusionScripts directly. In DaVinci Resolve, the Console is available by choosing Workspace > Console or choosing View > Console in Fusion Studio. There is also a Console button in the Fusion Studio User Interface toolbar.

Occasionally the Status bar displays a badge to let you know there's a message in the Console you might be interested in.



The Console window

A toolbar at the top of the Console contains controls governing what the Console shows. At the top left, the Clear Screen button clears the contents of the Console. The next four buttons toggle the visibility of error messages, log messages, script messages, and input echoing. Showing only a particular kind of message can help you find what you're looking for when you're under the gun at 3:00 in the morning. The next three buttons let you choose the input script language. Lua 5.1 is the default and is installed with Fusion. Python 2.x and Python 3.x require that you install the appropriate Python environment on your computer. Because scripts in the Console are executed immediately, you can switch between input languages at any time.

At the bottom of the Console is an Entry field. You can type scripting commands here for execution in the current comp context. Scripts are entered one line at a time, and are executed immediately. For more information on scripting, see the Fusion Scripting Manual.

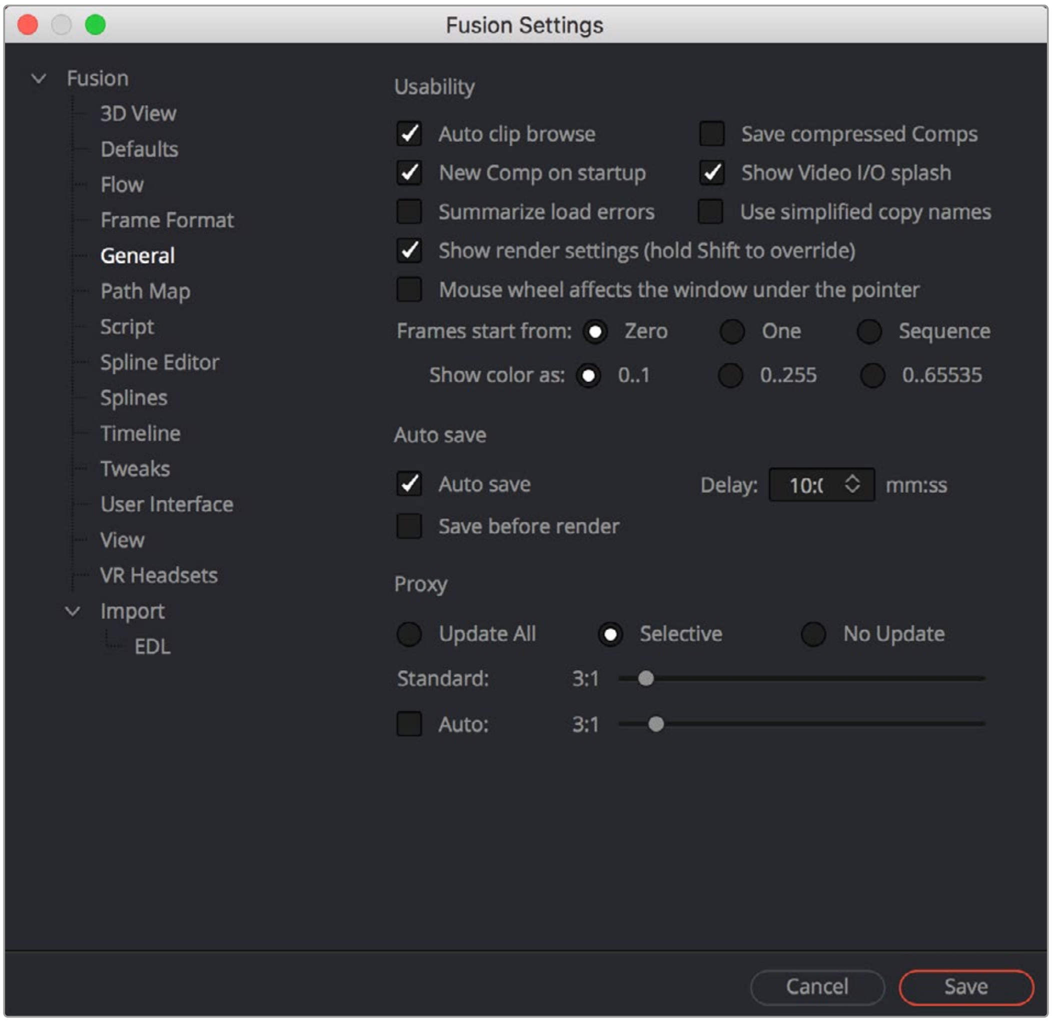
Customizing Fusion

This section explains how you can customize Fusion to accommodate whatever workflow you're pursuing.

The Fusion Settings Window

Fusion has its own settings window, accessible by choosing Fusion > Fusion Settings in DaVinci Resolve, or in Fusion Studio by choosing Fusion > Preferences on macOS or File > Preferences on Windows or Linux. This window has a variety of options for customizing the Fusion experience.

For more information, *see Chapter 75, "Preferences,"* in the DaVinci Resolve Reference Manual or Chapter 15 in the Fusion Reference Manual.



The Fusion Settings window set to the General panel

FUSION

Saving Fusion Layouts

It is possible to customize the layout and configuration of panels to suit the size of the desktop and monitor, or to match personal preferences.

In DaVinci Resolve, configure and resize the panels you want displayed and then:

Choose Workspace > Layout Presets > Save Layout Presets.

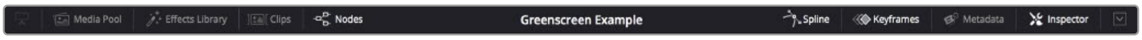
In Fusion Studio, configure and resize the panels you want displayed and then:

Click the Grab Document Layout button in the Preferences > Layout panel to save the layout for all new Compositions.

Click the Grab Program Layout button to remember the size and position of any floating views, and enable the Create Floating Views checkbox to automatically create the floating windows when Fusion restarts.

Showing and Hiding Panels

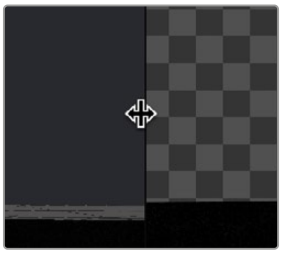
The UI toolbar at the top of the screen lets you open panels you need and hide those you don't. It's the simplest way to create a layout for your particular needs at the moment.



The UI toolbar of the Fusion page

Resizing Panels

You can change the overall size of each panel using preset configurations, or you can adjust them manually. The viewers and Work panel are inverse of each other. The more space used to display the Work panel, the less space available for the viewers. To resize a panel, manually drag anywhere along the raised border surrounding the edges of the panel.



Dragging the edge between two viewers to resize it

Fusion Studio Floating Frame

Fusion Studio includes a Floating Frame window that can be used to house any panel.

To place a panel in the Floating Frame, do the following:

1. In Fusion Studio, choose Window > New Floating Frame.
2. Right-click in the Floating Frame and choose the panel from Add View submenu.

When using multiple monitors, you can choose to have floating panels spread across your displays for greater flexibility.

Fusion Keyboard Remapping

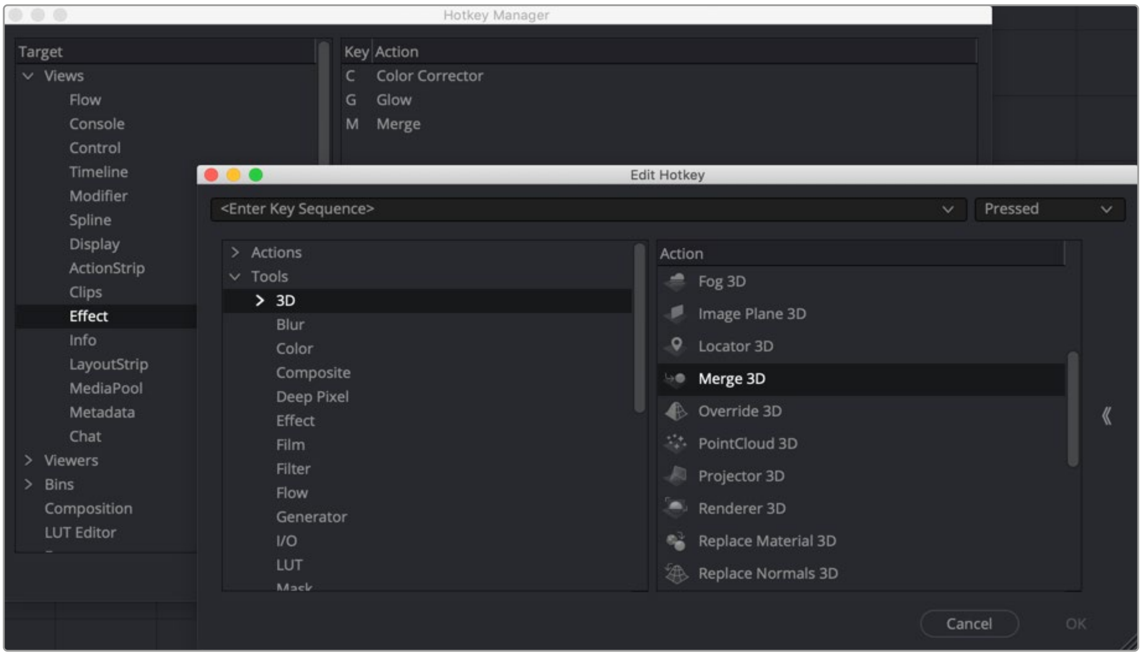
When using the Fusion page, functions and tools can be mapped to hot keys on your keyboard by choosing DaVinci Resolve > Keyboard Customization. In the Commands column of the Keyboard Customization window, select Panels > Fusion Page.

For more information on using the Keyboard Customization window, *see Chapter 4, "System and User Preferences."*

Fusion Studio Keyboard Remapping

When using Fusion Studio, functions and tools can be mapped to hot keys on your keyboard by choosing Views > Customize Hotkeys.

The Fusion Hotkey Manager dialog is divided into two sections. The left is where you select the functional area where you want to assign a keyboard shortcut. The right side displays the keyboard shortcut if one exists. You can use the New button at the bottom of the dialog to add a new keyboard shortcut.



The Fusion Studio Hotkey Manager window

For instance, if you want to add a shortcut for a specific node:

1. Open the Keyboard Hotkey Manager.
2. Select Views > Effect from the Target area of the Hotkey Manager.
3. Below the Key/Action area, click the New button to create a new keyboard shortcut for the Node.
4. In the Edit Hotkey window, click the Tools disclosure arrow, and then select Blur to display all the Blur-related nodes.

5 Select Glow in the Action panel.
6 At the top of the Edit Hotkey window, type G as the shortcut for the Glow node, and then click OK.
7 Glow and the G hotkey will now appear in the Key/Action area on the right.
8 Click OK to close the Hotkey Manager.
9 Click in the Node Editor and press G to add a Glow node.

Undo and Redo

Undo and Redo commands let you back out of steps you've taken or commands you've executed and reapply them if you change your mind. Fusion is capable of undoing the entire history of things you've done since creating or opening a particular project. When you close a project, its entire undo history is purged. The next time you begin work on a project, its undo history starts anew.

There is no practical limit to the number of steps that are undoable (although there may be a limit to what you can remember).

To simply undo or redo changes you've made one at a time:

Choose Edit > Undo (Command-Z) to undo the previous change.

Choose Edit > Redo (Shift-Command-Z) to redo to the next change.

Chapter 65

Getting Clips into Fusion

This chapter details the various ways you can move clips into Fusion as you build your compositions.

Contents

FUSION